

Chapter 4 :

Bipolar Junction Transistor (BJT)

4.1 Introduction

During the period of 1904 to 1947, vacuum tubes were the only electronic devices to build electronic circuits and systems. On December 24, 1947, the revolutionary electronic device **transistor** was invented by Dr. S. William Shockley, Walter H. Brattain, and John Bardeen at the Bell Telephone Laboratories. Since then, this device has changed the history of electronics industries. Compared to vacuum tubes, transistors are very small and efficient because it does not require any heating mechanism as vacuum tubes do. Even today, transistors are used in developing electronic circuits both in discrete form and in integrated circuit (IC) form. This chapter of this book will cover the construction, operation, and characteristics of bipolar junction transistor (BJT). Another transistor called **field effect transistor** (FET), which is widely used in IC fabrications, will be discussed in **Chapter 5**. The name transistor comes from two words- "**transfer**" and "**resistor**" = "**transistor**". This device transfers a signal from a low-resistance circuit to a high-resistance circuit. For this reason, it is called "**transfer-resistor**" or "**transistor**".

4.2 Construction of a Transistor

A bipolar junction transistor consists of two back-to-back PN junctions manufactured in a single piece of semiconductor material crystal. It can be formed from a bar of N-type semiconductor crystal that has been doped in such a way that it changes from N to P and back to N, that is, sandwiching a P layer in between two N-layers as shown in Fig.4-1. This results in what is called an NPN transistor. Alternately, transistor can be manufactured by sandwiching an N-layer in between two P-layers as in Fig.4-2. This configuration is called a PNP transistor.

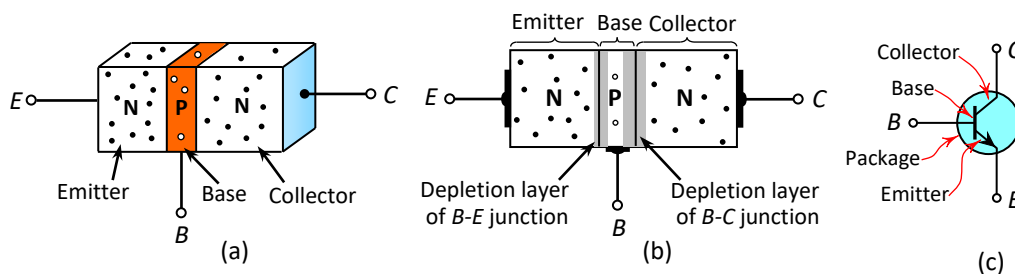


Fig.4-1: Construction of NPN transistors, (a) Three-dimensional block, (b) Two-dimensional block, and (c) Symbol of an NPN transistor

The middle layer of a transistor is called the **base** (*B*), whereas, of the remaining two layers one is called the **emitter** (*E*) and the other is called the **collector** (*C*). Conducting terminals are attached to each layer of the transistor so that external electrical connections can be made with them. The emitter layer is heavily doped, while the base and collector layers are only lightly doped. The outer layers have widths much greater than the sandwiched P or N layer. The ratio of the total device-

width to that of the center layer is 150:1. Depending on the application, i.e., required current capacity and operating frequency the actual dimension may vary from 10 to several 10s of microns. The doping of the sandwiched layer is also considerably less than that of the outer layers (typically, 1:10 or less).

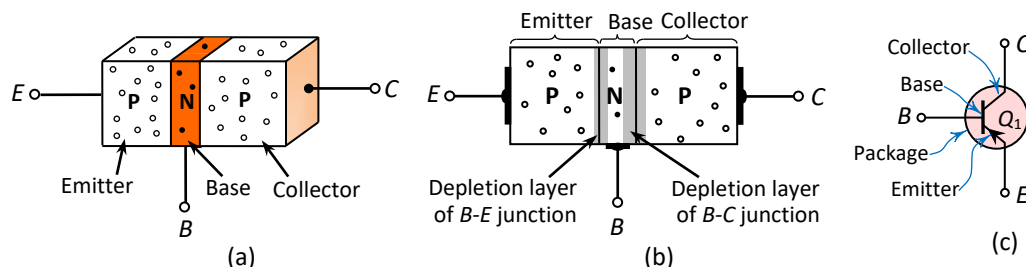


Fig.4-2: Construction of PNP transistors, (a) Three dimensional block, (b) Two dimensional block, and (c) Symbol of a PNP transistor

Look at the symbols of the transistors. The base layer is in the middle and the collector and the emitter are connected to the base. Emitter is differentiated from the collector by an arrow. In NPN transistor the arrow direction is outward from the base and in PNP transistor the direction of the arrow is towards to base. This arrow direction indicates the direction of conventional current flow. These types of transistors contain two PN junctions ($B-E$, and $B-C$), and both the majority and minority charge carriers (electrons and holes) constitutes the flow of currents. For this reason, they are called **Bipolar Junction Transistor (BJT)**.

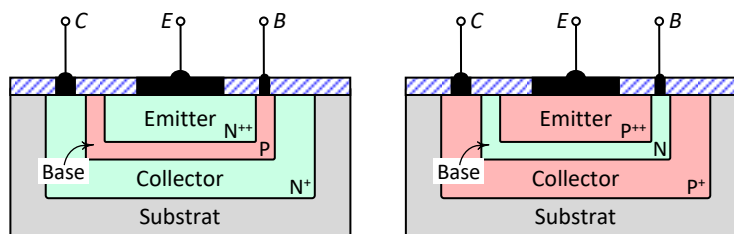


Fig.4-3: Epitaxial grown BJT: (a) NPN transistor and (b) PNP

The figures 4-1 and 4-2 show the constructional blocks of NPN and PNP transistors, respectively. In industries, transistors are constructed using some other different methods. The cross-sectional views of an NPN and a PNP transistor are shown in Fig.4-3. The junctions are, in fact, produced in a single slice of silicon (called substrate) by diffusing impurities through a photographically reduced mask. The construction process is mainly discussed in VLSI and IC fabrication courses.

4.3 Operation of a Transistor

The operation of a transistor will be described using an NPN transistor of Fig.4-1(a). The operation of the PNP transistor is exactly the same except holes will be the majority carriers and electrons will be minority carriers and the current directions and biasing voltage will be just reverse. For proper operation of a transistor the base-emitter junction ($B-E$ junction) must be forward biased and the base-collector junction ($B-C$ junction) must be reverse biased as shown in Fig.4-4. Look at the Fig.4-

4 the negative voltage to the emitter and the positive voltage to the collector have been supplied with respect to the base. Or the base is common to both the input circuit ($B-E$ circuit) and the output circuit ($B-C$ circuit). For this reason this configuration is called **Common Base Configuration (CBC)**.

Due to the forward bias, the depletion region of $B-E$ junction is reduced, that results in a heavy flow of majority carriers electrons from the N-type emitter to the P-type base material. These injected electrons are the minority carriers in P-type base. Due to the reverse bias, the depletion layer of the $B-C$ junction is widened.

As we know the electric field of the reverse biased PN junction assists the flow of minority carriers, most of the electrons injected by the emitter (which are the minority carriers in base) manage to reach the collector layer. As the base is lightly doped and its width is very small, only a few of the electrons recombine with the majority holes in the base layer. For example, if 100

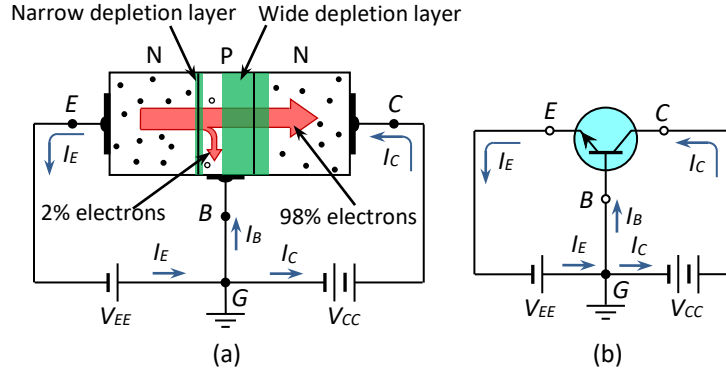


Fig.4-4: Operation of an NPN transistor, (a) Circuit with transistor block, and (b) Circuit with transistor symbol

electrons are injected by the emitter per second, less than 2 electrons will recombine in the base and more than 98 will move to the collector. As the direction of conventional current is just opposite to the direction of electron flow, a current will flow from the emitter to battery, which is called emitter current, I_E . Due to the recombination of 2 holes (by 2 electrons) in the base-region, a small current will flow into the base and this current is called base current, I_B . On the other hand, 98 electrons that move to the collector constitute a current which is called injected collector current $I_{C(INJ)}$ and its direction is from the positive terminal of battery to the collector as shown in Fig.4-4. Now, if we apply KCL at point G, we have,

$$I_E = I_{C(INJ)} + I_B \quad (4-1)$$

From this operation, it is now clear that more than 98% of emitter current is transferred to collector and only less than 2% is lost in the base. This current transfer action is called the **transistor action**. Due to the forward bias, $B-E$ circuit has very low resistance, and due to the reverse bias the $B-C$ circuit has very high resistance. But the transistor has transferred the current from the low resistance circuit to the high resistance circuit. That is why this device is called "**transfer-resistor**" or "**transistor**". **The higher the emitter current is transferred to the collector the better is the transistor action.** The ratio of the injected collector current to the emitter current is called the **current gain** (α) of a transistor in common base configuration (CBC). That is,

$$\alpha = \frac{I_{C(INJ)}}{I_E} \quad (4-2)$$

or,

$$I_{C(INJ)} = \alpha I_E \quad (4-3)$$

The value of α is less than unity and ranges from 0.95 to 0.997.

The $I_{C(INJ)}$ is not only the current component in the collector. We know that in every reverse biased P-N junction a leakage current flows due to the thermally generated minority carriers. For the same reason, a very small leakage current will flow through the reverse biased $B-C$ junction. As shown in Fig.4-5, if $I_E = 0$, that is, if the emitter terminal is open, a leakage current will flow through the collector due to the minority carriers. This current is called **Collector to Base current with emitter Open** and is denoted as I_{CBO} . In transistor data sheet, this current is also called **collector cutoff current** or **leakage current**. Although, this current is very small and becomes saturated for a small reverse bias, it increases with temperature (doubles for every 10°C rise in temperature). Thus, the collector current I_C will have two components as,

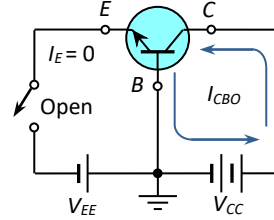


Fig.4-5: Collector leakage current with emitter open

$$I_C = I_{C(INJ)} + I_{CBO} \quad (4-4)$$

or,

$$I_{C(INJ)} = I_C - I_{CBO}$$

$\therefore$ from Equ.(4-2),

$$\alpha = \frac{I_{C(INJ)}}{I_E} = \frac{I_C - I_{CBO}}{I_E} \quad (4-5)$$

But, as the value of I_{CBO} is very small it can be neglected. That is $I_C \approx I_{C(INJ)}$. Therefore we can write,

$$\alpha \approx \frac{I_C}{I_E} \quad (4-6)$$

Using Equ.(4-2) and (4-4), we can also write,

$$I_C = \alpha I_E + I_{CBO} \quad (4-7)$$

Using this approximation ($I_C \approx I_{C(INJ)}$), Equ.(4-1) can be written as,

$$I_E = I_C + I_B \quad (4-8)$$

Although, we have found that the injected collector current $I_{C(INJ)} (= \alpha I_E)$ moves through the $B-C$ junction for the same mechanism as the minority carriers (attraction of the reverse biased electric field), this current is called the majority carriers current, as $I_{C(INJ)}$ is very much greater than the 2nd current component I_{CBO} . In fact, $I_{C(INJ)}$ comes from the majority carriers flow through the forward biased $B-E$ junction. Current component I_{CBO} is called the minority current component as it is thermally generated and quite smaller than $I_{C(INJ)}$. Thus we can say,

$$I_C = I_{C(majority)} + I_{C(minority)} = I_{C(INJ)} + I_{CBO} \quad (4-9)$$

Since, in a transistor current flow takes place for the movement of both the majority carriers and the minority carriers, it is called **Bipolar Junction Transistor** or BJT.

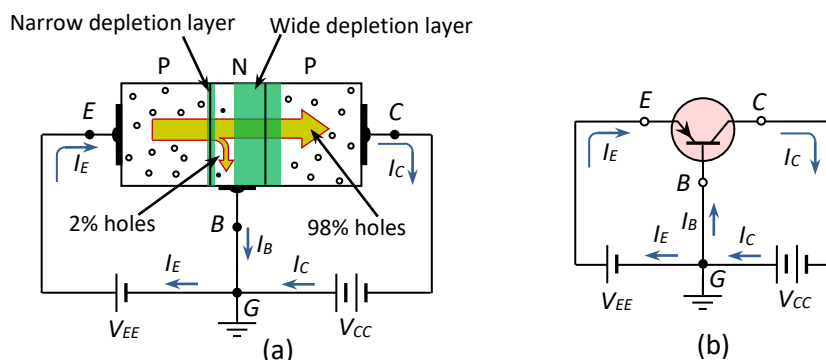


Fig.4-6: Operation of an NPN transistor, (a) Circuit with transistor block, and (b) Circuit with transistor symbol

Operation of a PNP transistor is exactly same, except here holes move from emitter to collector through base, instead of electrons. The biasing voltages of the PNP transistor will also be in reverse direction as shown in Fig.4-5.

Example 4-1

A copper penny has a mass of 4 gm. If all the free electrons of the penny flow uniformly through the emitter of a transistor in a whole day, calculate the three current components in the transistor. The mass number of Cu is 64 and Avogadro's number is 6×10^{24} , charge of an electron is $q = 1.6 \times 10^{-19}$ Coulombs, and the value of α of the transistor is 0.95.

Solution:

First, we have to calculate the number of atoms in a copper penny using the following formula,

$$\frac{N}{N_0} = \frac{m}{M}$$

$$\text{or, } N = \frac{m \times N_0}{M} = \frac{(4 \text{ gm})(6 \times 10^{24} \text{ atoms/mole})}{64 \text{ gm/mole}} = 2.814 \times 10^{22} \text{ atoms} \quad (4-10)$$

We know that every copper atom gives one free electron. So the total charge of the free electron of the copper penny is,

$$Q = (2.814 \times 10^{22} \text{ atom}) \left(1 \frac{\text{electron}}{\text{atom}} \right) \left(1.6 \times 10^{-19} \frac{\text{Coulombs}}{\text{electron}} \right) = 4.5 \times 10^4 \text{ Coulombs}$$

Thus, the current in the emitter terminal will be,

$$I_E = \frac{Q}{t} = \frac{4.5 \times 10^4 \text{ Coulombs}}{24 \times 60 \times 60 \text{ s}} \approx 52.1 \text{ mA [Ans.]}$$

The value of the collector current will be (from Equ.4.5),

$$I_C \approx \alpha I_E = 0.95 \times 52.1 \text{ mA} \approx 49.5 \text{ mA [Ans.]}$$

Therefore, the value of base current will be (from Equ.4.7),

$$I_B = I_E - I_C = 52.1 \text{ mA} - 49.5 \text{ mA} = 2.6 \text{ mA [Ans.]}$$

Comments: Imagine the number of free electrons in Cu. Only a 4 gm Cu-penny can produce 52 mA of current for a whole day long.

4.4 Amplifying Action of a Transistor

The capacity of a transistor to transfer a signal from a low resistance circuit to a high resistance circuit is the key of a transistor for voltage amplification. As shown in Fig.4-6, even if we connect a large resistor R_C (say 2 k Ω) at the collector circuit, transistor action will not be hampered, i.e., 98% of the emitter current will be transferred to the collector. Now let us consider that the emitter bias voltage $V_{EE} = 1$ V and the forward resistance of the B-E junction is 50 Ω . So in the emitter circuit the value of I_E will be $(V_{EE} - V_{BE}) \div r_D = (1 \text{ V} - 0.7 \text{ V}) \div 50 \Omega = 6 \text{ mA}$. Due to the transistor action 98% of this current will be transferred to the collector and the value of I_C will be 5.88 mA. If we take the output voltage (v_o) across the external resistor R_C , according to Ohm's law its value will be $5.88 \text{ mA} \times 2 \text{ k}\Omega = 11.76 \text{ V}$! By applying 1 V in the input circuit we have found 11.76 V at the output. In this way, a transistor increases the input voltage level at the output circuit. Here, DC voltage has been amplified. Transistor is rarely used for DC voltage amplification.

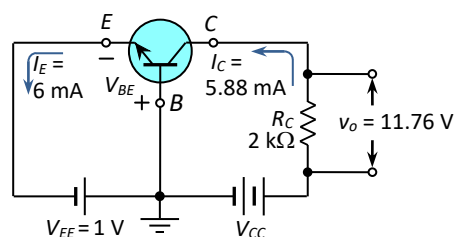


Fig.4-7: Amplifying action of transistor

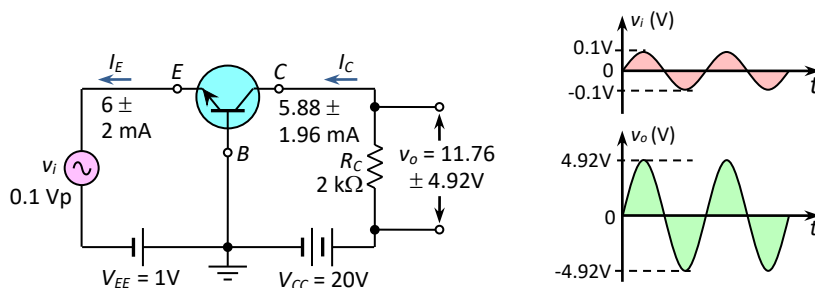


Fig.4-8: AC amplification using transistor

As shown in Fig.4-8, if we connect an AC source of very small voltage, say $v_i = 0.1 \sin \omega t$, in series with the biasing voltage V_{EE} , an AC current will also flow in the emitter circuit. In this case the peak value of the AC current will be $(0.1 \text{ V} \div 50 \Omega) = 2 \text{ mA}$. So, the peak value of the AC current in the collector circuit will be 1.96 mA ($2 \text{ mA} \times 98\%$). The AC peak voltage across the resistor will be 4.92 V ($1.96 \text{ mA} \times 2 \text{ k}\Omega$) as shown in the Fig.4-8. In this way, transistor can amplify AC voltage also. The output voltage will have a DC level, but here it is not shown.

Although, a transistor in CB configuration (as in Fig.4-7 and 4-8) provides voltage amplification, it cannot produce current amplification as the output current I_C is less than the input current I_E . However, there are some other configurations that can produce current amplification, and will be discussed here.

4.5 Current Directions and Voltage Notations in Transistors

In a transistor there are three terminals: *emitter (E)*, *collector (C)*, and *base (B)*. When properly biased, these three terminals carry three currents and potential differences are developed across them. As various types of voltage differences and currents are used to describe the performance and output of NPN and PNP transistor in different types of circuit, it would be wise to describe the current and voltage notations used for transistors. Fig.4-9 shows the current direction and the voltages used in an NPN transistor in common base configuration. Look at the current direction. The emitter current I_E will always be in the same direction as the arrow of the transistor symbol. In case of NPN transistor it will always be outward from the transistor irrespective of the circuit configuration. We know, $I_E = I_C + I_B$. So the direction of collector current I_C , and base current, I_B will be just opposite to that of I_E . In case of NPN transistor, they will flow into the transistor, no matter in what circuit the transistor is used.

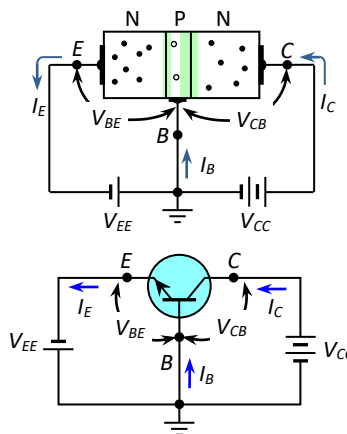


Fig.4-9: Voltage notations and current directions in an NPN

In case of a PNP transistor the emitter current I_E will enter into the transistor and the other two currents I_B and I_C will leave the transistor as shown in Fig.4-10.

To represent the potential difference (or simply voltage) across various nodes/terminals of a transistor circuit, **double subscript notations** are used. In this notation, the voltage difference is represented by V with the points across which V is measured is used as subscripts. The assumed higher potential point will be written first and then the lower potential or reference point. For example, in Fig.4-9, V_{BE} is used to denote the voltage difference between the base terminal (B) and the emitter terminal (E). Here, base has been assumed to be at higher potential than emitter, or the polarity of V_{BE} is assumed to be +ve at base and -ve at emitter. If the measured or calculated value of V_{BE} is positive, it matches the assumption, i.e., the base is at higher potential than the emitter. But, if the measured or the calculated value of V_{BE} is negative, it means that the base is at lower potential than the emitter. Similarly, V_{CB} represents the potential difference between the collector and the base. Here, collector is assumed to be at higher potential than the base. The assumed polarity of V_{BE} and V_{CB} ensure the forward bias of B-E junction and the reverse bias of the B-C junction.

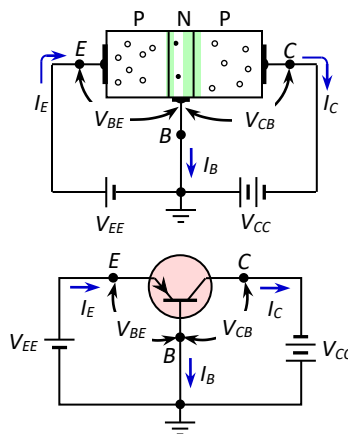


Fig.4-10: Voltage notations and current directions in a PNP

If a voltage is measured at any point of a transistor with respect to ground, for example at the collector, it should be V_{CG} . But the G subscript is not used. Hence, it will be simply V_C . Similarly the voltage at base and emitter with respect to ground will be V_B and V_E , respectively.

The batteries (voltage sources) or power supplies are normally connected to different terminals of a transistor with respect to ground, i.e., one terminals of the batteries are connected to the ground. For example, in Fig.4-9, the negative terminal of a small battery is connected to the emitter and the positive terminal is connected to ground. Here, also the letter G is not used, but the terminal to which it is supplying power is written twice. So for the battery connected to the emitter is represented by V_{EE} . Similarly, the battery connected to the collector is denoted by V_{CC} . If a battery is used to bias the base it will be represented by V_{BB} .

The polarity of the bias voltage of a PNP transistor will just be opposite to that of an NPN transistor. Now, the voltage between the base and emitter terminals should be denoted by V_{EB} as emitter is at higher potential. But in this text, exactly the same voltage notation will be used for both NPN and PNP transistors to avoid difficulties. Hence, for PNP transistor, instead of using V_{EB} and V_{BC} the same voltage notation of NPN transistor V_{BE} and V_{CB} will be used. Therefore, according to the rules describe above, voltages for NPN transistor will be positive and the voltages for PNP transistor will be negative as shown in Fig.4-10.

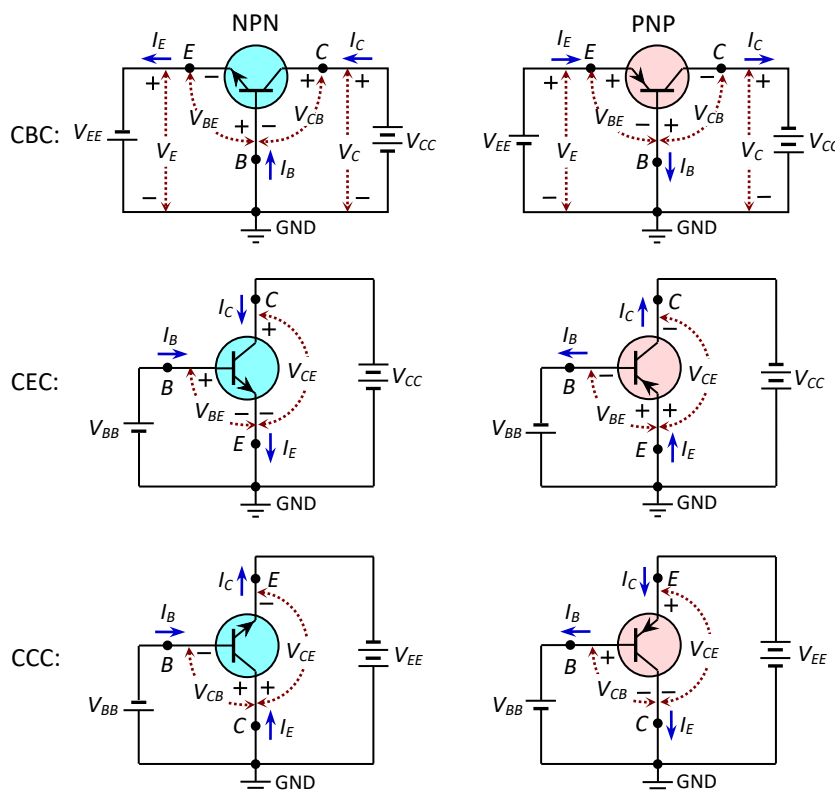


Fig.4-11: Current directions and voltage notations of different transistor circuit

To make the concept clearer, verify the rules discussed above in the circuits of Fig.4-11 (three configurations of transistor). Current directions are along the transistor arrow, and for NPN transistor higher potential letter is written first, and for PNP transistor lower potential terminal is written first. Carefully look at the circuits in CCC (common collector configuration) of Fig.4-11. Here, V_{CE} is positive for PNP transistor and negative for NPN transistor.

4.6 Characteristics of a Transistor

For any device, I - V characteristics are very important to understand the operation of that device. Although characteristic curves are seldom used in actual design or problem analysis, they convey a wealth of information. As a transistor has three terminals, keeping any one terminal common to the input and output circuits three configurations are possible. These are

- Common Base Configuration (CBC)
- Common Emitter Configuration (CEC) and
- Common Collector Configuration (CCC)

4.7 Characteristics of a Transistor in CBC

In common base configuration, the base is common to the input and output circuits. The emitter and base constitute the input circuit, and the base and collector constitute the output circuit. The base-emitter voltage (V_{BE}) is regarded as the input voltage and the emitter current (I_E) is regarded as the input current. Similarly, the base-collector voltage (V_{CB}) is the output voltage and the collector current (I_C) is the output current. We have to study the characteristics of both input and output circuits. Here, we will discuss them using an NPN transistor. The characteristics of PNP transistor will exactly be same but the direction of current will be just opposite and the polarity of biasing voltage will be reverse.

4.7.1 Input Characteristics

This characteristic shows the relation between the input current, I_E , and the input voltage, V_{BE} , for a fixed value of collector-base voltage (or output voltage) V_{CB} . To draw the input characteristic curve, the value of V_{BE} is gradually changed and the value of I_E is measure keeping a constant

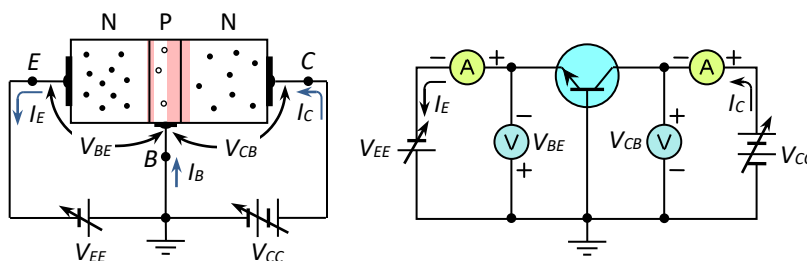


Fig.4-12: Circuit for determining transistor characteristics in CBC

value of V_{CB} using a circuit of Fig.4-12. [In some books, mainly practical books, series resistors are used in input and output circuits. But as the increase in I_E will also causes increase in I_C , the output voltage, V_{CB} , will be changed, due to the larger voltage drop across the series resistor. So in every step of data recording, V_{CC} has to be adjusted to keep V_{CB} constant.] By plotting the data on a

$V_{BE} - I_E$ coordinate system an input characteristic curve is found. The process is repeated for different (constant) levels of V_{CB} . All the graphs are plotted in the same coordinate system. This is called a **family of input characteristic curves**. A typical family of input characteristic curves for an NPN transistor is shown in Fig.4-13. Since the input is across the forward biased base-emitter junction, the graph of input current (I_E) versus input voltage (V_{BE}) has exactly the same shape as that of a forward biased PN junction. For a silicon transistor the value of I_E is almost zero up to $V_{BE} = 0.5$ V. After that I_E starts to increase and has a knee point at $V_{BE} \approx 0.7$ V. For a germanium transistor the value of knee voltage will be ≈ 0.3 V.

The shape of these $I_E - V_{BE}$ curves depends on the reverse bias output voltage V_{CB} . Fig.4-13 shows three graphs for three different values of V_{CB} ($= 20$ V, 10 V, and 2 V). Here, the input current increases with the increase in output voltage. As indicated by the horizontal lines, for a fixed value of V_{BE} (say 0.75 V) higher input current, I_E , is produced for higher values of V_{CB} . With the increase of the value of reverse bias voltage V_{CB} , the width of the $B-C$ depletion region increases. This reduces the width of the base region. With the decrease in base width its carrier concentration increases. Thus more electrons can diffuse from the emitter to the base which results in an increase in I_E . That's why I_E increases with the increase of V_{CB} . The decrease of base width with the increase of $B-C$ junction reverse bias is called **base width modulation** or **early effect**. The value of the DC input resistance of this forward biased $B-E$ junction is same as that of a forward biased PN junction diode, i.e., in the order of few Ohms (typically 50 to 100). AC resistance is even lower.

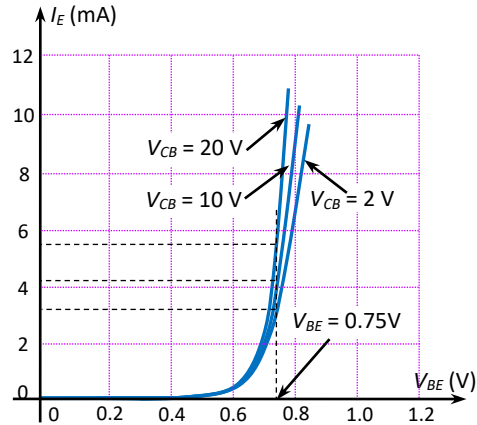


Fig.4-13: Common-base input characteristic (NPN)

Example 4-2

Calculate the DC resistance and AC resistance for $V_{BE} = 0.75$ V on the input characteristics of a transistor in CB configuration shown in Fig.4-14. Assume, $r_B = 2 \Omega$.

Solution:

First, we have to draw a vertical line on $V_{BE} = 0.75$ V (as shown in the figure). Then we have to find the intersecting point of this vertical line with the I - V characteristics. Now, if we draw a horizontal line through this point it will intersect the y -axis. This intersecting point gives the value of I_E , which is ≈ 6.4 mA.

Now, using Equ.(2-37) of Chapter 2, the value of DC resistance is,

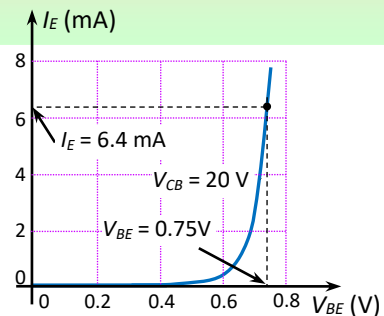


Fig.4-14: Input graphs for example 4.

$$R_D = \frac{V_D}{I_D} = \frac{0.75 \text{ V}}{6.4 \text{ mA}} \approx 117 \Omega \text{ (Ans.)}$$

Using Equ.(2-42), the value of AC resistance is,

$$r_d = \frac{26 \text{ mV}}{I_D} + r_B = \frac{26 \text{ mV}}{6.4 \text{ mA}} + 2 \Omega \approx 6 \Omega \text{ (Ans.)}$$

Comments: The resistance of the input circuit is similar to that of the forward biased PN junction diode.

4.7.2 Output characteristics

The output characteristics can be determined using the same schematic diagram of Fig.4-12. For this purpose, the collector (output) current, I_C , is measured as the output voltage V_{CB} is adjusted, for a fixed value of input (emitter) current I_E . When the measured data are plotted on a I_C versus V_{CB} coordinate system, we get an output characteristic curve. By repeating the process for different fixed values of I_E we can draw a family of output characteristic curves as shown in Fig.4-15.

We observe, first, in this figure that each curve starts from the left side of $V_{CB} = 0$ line and increases rapidly just as V_{CB} increases slightly from its negative value. At $V_{CB} = 0$ I_C becomes almost equal to I_E and the ratio gives the value of current gain in CBC ($\alpha = I_C / I_E$). Afterwards, the curves become almost parallel to x -axis which means that I_C increases by a very small amount with the increase in output voltage, V_{CB} . The minority carriers injected into the base (by the emitter) are swept in by the in-built electric field of the B - C junction, even when $V_{CB} = 0$. For this reason, I_C becomes almost equal to I_E when $V_{CB} = 0$. We know that, minority carriers cannot move through a forward biased PN junction. Hence, for a little forward bias of B - C junction (negative value of V_{CB}), I_C falls to zero. That is why the curves start from the right side of $V_{CB} = 0$ line. Again, with the increase of V_{CB} the base width decreases (base width modulation) and I_C increases by a very small amount. At this region, I_C can be changed by changing the value of I_E (as shown in Fig.4-15).

This family of output characteristic curve has three regions. These are: **saturation region**, **cutoff region**, and **active region**.

The region, where the value of I_C increases with the increase of V_{CB} is called the **saturation region**. Certainly, for this common-base configuration, this region lies to the right of $V_{CB} = 0$ line. Here, the -ve values of V_{CB} imply that the B - C junction is not reverse biased, rather it is forward biased. So, we find that in this region both the B - E junction and the B - C junctions are forward biased. Transistors are used in this region only when they work as a switch (as an ON switch).

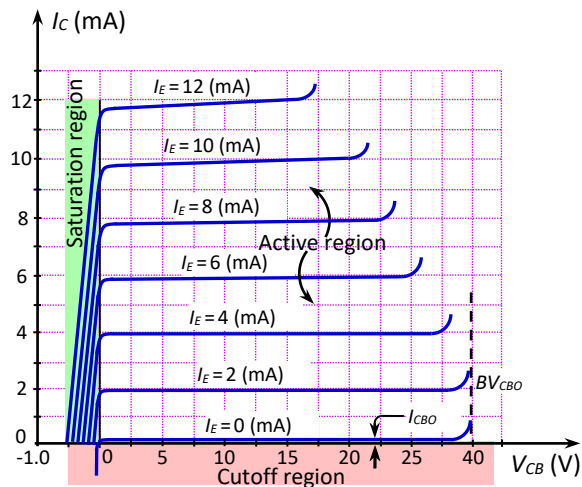


Fig.4-15: Common-base output characteristics (NPN)

As we know, when the emitter terminal is open, i.e., when $I_E = 0$, a very small current I_{CBO} flows through the collector [Fig.4-5]. On the output characteristic curve this line passes almost along the x -axis. The region below this curve is called the **cutoff region**. As no bias voltage is applied to the B - E junction (emitter is open) and the B - C junction is reverse biased by V_{CB} , we can say in this cutoff region both the junctions are reverse biased. This region is also used for switching purpose (OFF switch).

The region in between the saturation and the cutoff regions is called the **active region**. In this region, the B - E junction is forward biased (as $I_E > 0$) and the B - C junction is reverse biased by V_{CB} . This is the region where a transistor works as an amplifier. Look at the shape of the curves in this region: with a very large change in the output voltage V_{CB} , practically no change is found in I_C . This ensures a very large output resistance of the transistor when works in active region ($r_o \approx \infty$).

In addition to these three regions, there is another region where we should not operate a transistor, because in this region the transistor breaks down. As is the case of a reverse biased diode, the current through the collector-base junction of a transistor may increase suddenly if the reverse bias voltage V_{CB} across it is made extremely large. This increase in current is mainly caused by the avalanche breakdown mechanism as described in Zener diode section in Chapter 2. However, in a transistor this sudden increase in I_C may be due to another phenomenon called **punch through** or **reach through**. Punch through occurs when the reverse bias widens the B - C depletion region to the extent that it meets the B - E depletion region. This joining of the two depletion regions effectively shorts the collector to the emitter and causes a substantial current flow. Punch through is a main design factor in determining the doping level and width of the base layer of a transistor. In Fig.4-15 at the end of each curve upward bending represents the starting to breakdown. The collector-to-base breakdown voltage when $I_E = 0$ (i.e., emitter is open) is designated as **Collector to Base Breakdown Voltage with emitter Open, BV_{CBO}** . Breakdown occurs progressively lower voltage for increasing values of I_E .

Although, normally a B - E junction is not reverse biased, in practical application it may be reverse biased by an AC signal. Of course, it may also breakdown, and remember that its reverse breakdown voltage is usually much less than BV_{CBO} .

4.7.3 Transfer Characteristics

The common-base transfer characteristics show the relations between the output current, I_C , and the input current, I_E , for a constant value of output voltage, V_{CB} . These are the graphs of I_C versus I_E for different fixed values of V_{CB} . These characteristics are also called **forward transfer characteristics** or **current gain characteristics**. These characteristics can be derived from the output characteristic curves or can be measured. To determine these graphs experimentally the same circuit of Fig.4-12 can be used. Keeping V_{CE} fixed at a suitable value, input current I_E is adjusted step by step. At every steps of I_E , the value of I_C is measured. The data are plotted on a $I_C - I_E$ coordinate system and a transfer

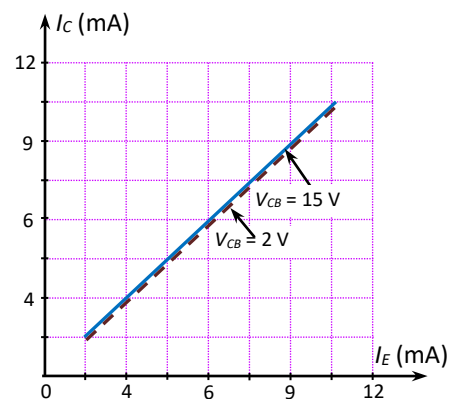


Fig.4-16: Transfer characteristic curves for CB configuration transistor characteristics in CBC

characteristics in CBC

characteristic curve is found. The process is repeated by fixing value of V_{CE} at different levels. In this way two or three curves are drawn. A set of typical transfer characteristic curves is shown in Fig.4-16. From this graph, it is clear that there is a linear relationship between I_C and I_E (which depends on α). The slope of this curve gives the value of α . For the variation of output voltage V_{CB} there is no significant change in this characteristic curve.

4.8 Current Gain in CBC

Although in Fig.4-15 it seems that the output current I_C is equal to the input current I_E , but actually I_C is always less than I_E . The ratio of the output current to the input current is called the current gain and is denoted by α (alpha). As the transistor can amplify both DC signal and AC signal (as described in Section 4.4) α can be defined separately.

4.8.1 Alpha in DC Mode (α_{dc})

When a DC voltage source is applied to the input circuit, an input current I_E and an output current I_C are produced (I_{CBO} has been neglected). I_C sets a fixed level of current at the output. The ratio of this fixed output current to the input current is called DC current gain α_{dc} . Mathematically,

$$\alpha_{dc} = \frac{\text{DC output current}}{\text{DC input current}} = \frac{I_C}{I_E} \quad (4-11)$$

Already we have found that the value of α_{dc} ranges from 0.98 to 0.997.

4.8.2 Alpha in AC Mode (α_{ac})

In addition to bias voltage, if an AC signal is added in the input circuit, an AC current is developed both in the input and output circuit. Due to the AC signal the input current and the output current levels oscillates (by a small amount) around a DC level. The ratio of the change in input current, ΔI_E , to the change in output current, ΔI_C , for a constant value of output voltage, V_{CB} , is referred to AC current gain α_{ac} . Mathematically,

$$\alpha_{ac} = \frac{\text{Change in output current}}{\text{Change in input current}} \bigg|_{V_{CB}(\text{constant})}$$

or,

$$\alpha_{ac} = \frac{\Delta I_C}{\Delta I_E} \bigg|_{V_{CB}(\text{constant})} \quad (4-12)$$

The AC alpha is also called the **common-base short-circuit amplification factor**. For most of the situations, the magnitudes of α_{ac} and α_{dc} are quite close to each other.

Example 4-3

From the output characteristic curves of a CB configuration transistor, given in Fig. 4.17, calculate the value of α_{dc} for $V_{CB} = 5$ V and $V_{CB} = 15$ V and $I_E = 6$ mA.

Solution:

To calculate the values of I_C , we have to draw two vertical lines on $V_{CB} = 5$ V and $V_{CB} = 15$ V (as shown on the figure). Then we have to find the intersecting points of these vertical lines with the

$I_E = 6$ mA graph. Now, if we draw two horizontal lines through these points they will intersect the y-axis. These intersecting points give the values of I_C corresponding on $V_{CB} = 5$ V and $V_{CB} = 15$ V. From the figure, we see that for $V_{CB} = 15$ V, $I_C \approx 6$ mA. For $V_{CB} = 5$ V, $I_C \approx 5.96$ mA that can be measured by expanding the graph. Therefore,

$$\text{When } V_{CE} = 5 \text{ V, } \alpha_{dc} = \frac{I_C}{I_E} \approx \frac{6 \text{ mA}}{6 \text{ mA}} = 1.0 \text{ [Ans.]}$$

$$\text{When } V_{CE} = 15 \text{ V, } \alpha_{dc} = \frac{5.96 \text{ mA}}{6 \text{ mA}} = 0.994 \text{ [Ans.]}$$

Comments: The value of α is approximately 1.

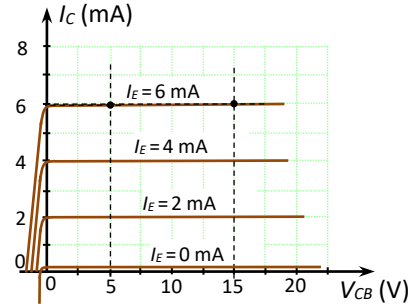


Fig.4-17: Graph for example 4

4.9 Characteristics of a Transistor in CEC

In common emitter configuration (CEC), the emitter is common to both input circuit and output circuit as shown in Fig.4-18. The emitter and base constitute the input circuit while the emitter and collector constitute the output circuit. The base-emitter voltage, V_{BE} , is regarded as the input voltage and the base current, I_B , is regarded as the input current. On the other hand, collector-emitter voltage, V_{CE} , is the output voltage and the collector current, I_C , is the output current. For proper understanding of the input and output characteristics of this configuration we have to describe some parameters first.

4.9.1 Current Gain Beta (β)

Unlike the CB configuration CE configuration gives high current amplification and it is defined as the ratio of the output current, I_C , to the input current, I_B , and is denoted by beta (β). It can also be measured for both DC current and AC current.

4.9.2 Beta in DC Mode (β_{dc})

When a DC voltage source is connected in the input circuit, an input current, I_B , is produced. Due to the transistor action, an output current, I_C , is also produced in proportion with I_B . I_C sets a fixed level of current at the output. The ratio of this fixed output current, I_C , to the input current, I_B , is called DC current gain, β_{dc} . Mathematically,

$$\beta_{dc} = \frac{\text{DC output current}}{\text{DC input current}} = \frac{I_C}{I_B} \quad (4-13)$$

The value of β_{dc} is much greater than unity and depends on the value of α_{dc} .

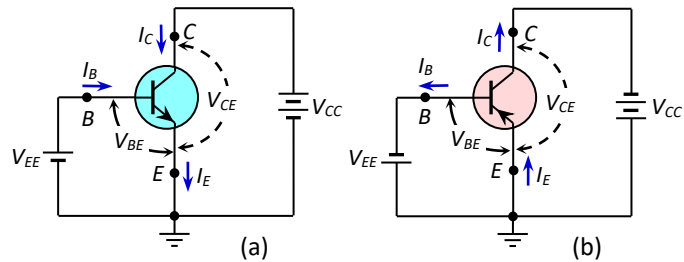


Fig.4-18: Common emitter configuration circuits

(a) NPN (b) PNP

4.9.3 Beta in AC Mode (β_{ac})

In addition to the DC bias voltage, if an AC voltage source is connected to the input circuit, an AC current is developed both in the input and output circuits. Due to the AC component the input and the output current levels oscillate (by a small amount) around a DC level. The ratio of the change in input current, Δi_B , to the change in output current, Δi_C , for a constant value of output voltage, V_{CE} , is referred to AC current gain β_{ac} . Mathematically,

$$\beta_{ac} = \frac{\text{Change in output current}}{\text{Change in input current}} \bigg|_{v_{CE}(\text{constant})} = \frac{\Delta i_C}{\Delta i_B} \bigg|_{v_{CE}(\text{constant})} \quad (4-14)$$

β_{ac} is also called **common-emitter forward-current amplification factor**. In specification sheets β_{dc} is normally referred to as h_{FE} and β_{ac} is referred to as h_{fe} .

Example 4-4

A properly biased NPN transistor in CEC, has a base current of $10 \mu\text{A}$. The current gain of a transistor is 120 (β_{dc}). Calculate the values of collector current and emitter current.

Solution:

Using Equ.(4-13), $I_C = I_B \times \beta_{dc} = 10 \mu\text{A} \times 120 = 1.2 \text{ mA}$ [Ans.]

Using Equ.(4-8), $I_E = I_B + I_C = 0.010 \text{ mA} + 1.2 \text{ mA}$ [Ans.]

Comments: To calculate we can also use this equation: $I_E = (\beta + 1)I_B$.

4.9.4 Variation of β_{dc}

β_{dc} or h_{FE} is a very important parameter of a bipolar junction transistor. For designing different circuits we must know about the β_{dc} of the transistor. Although, in most of the design, we consider β_{dc} as a constant parameter, but actually its value changes with the variation of operating temperature and collector current.

Keeping the junction temperature constant, β_{dc} increases with the increasing in I_C up to a maximum level. After that, β_{dc} decreases with the increasing in I_C as shown in Fig.4-18. On the other hand, if I_C is kept constant, and the temperature is varied, β_{dc} increases with the increase in temperature and vice versa. Fig.4-19 shows the variation of β_{dc} with I_C and junction temperature (T_j) for a typical BJT.

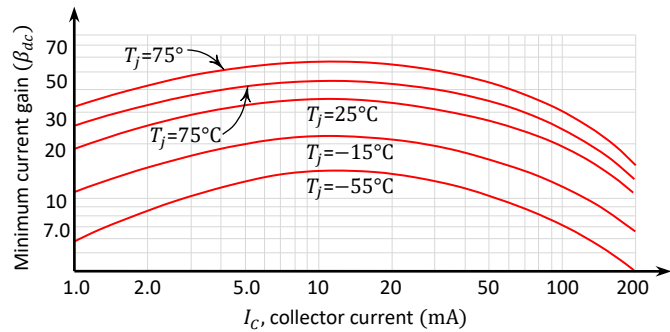


Fig.4-19: Variation of β_{dc} with collector current and temperature

A transistor datasheet usually specifies the value of β_{dc} (or h_{FE}) at specific values of I_C . Even at fixed values of I_C and temperature, β_{dc} varies from one device to another for a given type (number) of transistor. This

variation is due to the inconsistencies in the manufacturing process. The manufacturers provide the minimum value of β_{dc} at a certain value of I_C in the datasheets. However, some datasheets contain three value of β_{dc} : Minimum, typical and maximum values.

4.9.5 Relation Between α and β

The current gain in CEC (β) is directly related to the current gain in CBC (α). Although β_{dc} is related to α_{dc} and β_{ac} is related to α_{ac} , to make the expression easy here a generic relation is derived. From the discussion of transistor action we know [using Equ.(4-8)]

$$I_E = I_C + I_B$$

Dividing both side of this equation by I_C , we get

$$\frac{I_E}{I_C} = 1 + \frac{I_B}{I_C}$$

$$\text{or,} \quad \frac{1}{I_C/I_E} = 1 + \frac{1}{I_C/I_B}$$

$$\text{or,} \quad \frac{1}{\alpha} = 1 + \frac{1}{\beta} \quad [\because \frac{I_C}{I_E} \approx \alpha \text{ and } \frac{I_C}{I_B} = \beta]$$

$$\text{or,} \quad \frac{1}{\alpha} - 1 = \frac{1}{\beta}$$

$$\text{or,} \quad \frac{1 - \alpha}{\alpha} = \frac{1}{\beta}$$

$$\therefore \quad \beta = \frac{\alpha}{1 - \alpha} \quad (4-15)$$

$$\text{and,} \quad \alpha = \frac{\beta}{1 + \beta} \quad (4-16)$$

Example 4-5

An NPN transistor biased in CE configuration produces $I_B = 6 \mu\text{A}$, and $I_C = 1.08 \text{ mA}$. Calculate the value of I_E , β , and α .

Solution:

$$\text{Using Equ.(4-8),} \quad I_E = I_B + I_C = 0.006 \text{ mA} + 1.08 \text{ mA} = 1.086 \text{ mA} \text{ [Ans.]}$$

$$\text{Using Equ.(4-13),} \quad \beta = \frac{I_C}{I_B} = \frac{1.08 \text{ mA}}{0.006 \text{ mA}} = 180 \text{ [Ans.]}$$

$$\text{Using Equ.(4-16),} \quad \alpha = \frac{\beta}{1 + \beta} = \frac{180}{1 + 180} \approx 99.45 \text{ [Ans.]}$$

$$\text{Alternately,} \quad \alpha = \frac{I_C}{I_E} = \frac{1.08 \text{ mA}}{1.086 \text{ mA}} \approx 99.45 \text{ [Ans.]}$$

Comments: The value of α is almost 1. When the value of α increases, the value of β also increases.

4.9.6 Leakage Current in CEC (I_{CEO})

Like the CB configuration, CE configuration also produces a leakage current due to the thermally generated minority carriers. Fig.4-20 shows CE circuits where the base-emitter circuits are left open while the reverse biasing voltage sources remain connected at the collector circuits. The only current that can flow, when the base is left open, is the minority carriers reverse current through the C-B junction. This current flows from the collector, through the base region, and into the emitter. It is designated I_{CEO} —Collector to Emitter current with the base Open. Although, it is called ‘reverse’ current it flows in the same direction as the majority carriers collector current. In transistor data sheet, this current is also called *collector cutoff current* or *leakage current*.

4.9.7 Relation between I_{CEO} and I_{CBO}

I_{CEO} is directly related to the I_{CBO} . From previous discussion, we know [Equ.(4-7)]

$$I_C = \alpha I_E + I_{CBO}$$

$$\text{or, } I_C - I_{CBO} = \alpha I_E$$

Dividing both sides by α we get,

$$\frac{I_C}{\alpha} - \frac{I_{CBO}}{\alpha} = I_E$$

Substituting $I_B + I_C$ for I_E on the right-hand side,

$$\frac{I_C}{\alpha} - \frac{I_{CBO}}{\alpha} = I_B + I_C$$

$$\text{or, } I_C \left(\frac{1}{\alpha} - 1 \right) = I_B + \frac{I_{CBO}}{\alpha}$$

$$\text{or, } I_C \left(\frac{1 - \alpha}{\alpha} \right) = I_B + \frac{I_{CBO}}{\alpha}$$

Dividing both sides by $(1 - \alpha) / \alpha$,

$$I_C = \frac{\alpha I_B}{1 - \alpha} + \frac{I_{CBO}}{1 - \alpha} \quad (4-17)$$

But, when the base terminal is open, $I_B = 0$ and in that case $I_C = I_{CEO}$

Putting $I_B = 0$, and letting $I_C = I_{CEO}$, we get,

$$I_{CEO} = \frac{I_{CBO}}{1 - \alpha} \quad (4-18)$$

$$\text{or, } I_{CEO} = \frac{\alpha + 1 - \alpha}{1 - \alpha} I_{CBO} = \left(\frac{\alpha}{1 - \alpha} + \frac{1 - \alpha}{1 - \alpha} \right) I_{CBO} = (\beta + 1) I_{CBO}$$

$$\therefore I_{CEO} = (\beta + 1) I_{CBO} \quad (4-19)$$

If the value of $\alpha = 0.98$, from Equ.(4-18), we get, $I_{CEO} = I_{CBO} [1 / (1 - 0.98)] = 50 \times I_{CBO}$.

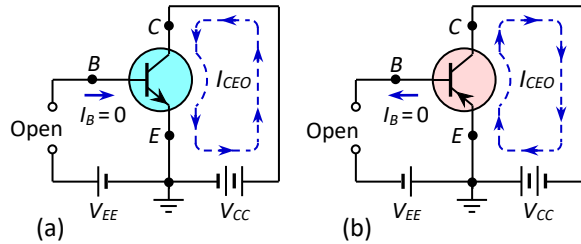


Fig.4-20: Leakage current in CEC (a) NPN, and (b) PNP transistor.

Thus, we find that the value of I_{CEO} is quite larger than the value of I_{CBO} .

Using equations (4-15), and (4-18) into Equ.(4-17) we get,

$$I_C = \beta I_B + I_{CEO} \quad (4-20)$$

Another useful relation is, $I_E = I_C + I_B$ [Equ. (4-8)] (4-21)

or, using Equ.(4-19), $I_E = \beta I_B + I_{CEO} + I_B$ (4-22)

Neglecting the leakage current I_{CEO} , $I_E = (\beta + 1)I_B$ (4-23)

Example 4-6

Applying appropriate biasing voltage in a PNP germanium transistor, two ammeters are connected in the collector and base. The readings of the meters are 60.20 mA and 0.4 mA, respectively. If the value of β of the transistor is 150, calculate the leakage currents, I_{CEO} and I_{CBO} .

Solution:

The reading of the two ammeters are 61 mA and 0.4 mA. First one gives the value of I_C and the second one gives the value of I_B , that is, $I_C = 60.20$ mA and $I_B = 0.4$ mA.

Using Equ.(4-20), $I_C = \beta I_B + I_{CEO}$

or, $I_{CEO} = I_C - \beta I_B$

or, $I_{CEO} = 60.20 \text{ mA} - 150 \times 0.4 \text{ mA} = 0.20 \text{ mA}$ [Ans.]

Again we know that, $I_{CEO} = (\beta + 1)I_{CBO}$ [Equ.(4-19)]

or, $I_{CBO} = \frac{I_{CEO}}{\beta + 1} = \frac{0.20 \text{ mA}}{150 + 1} \approx 1.33 \mu\text{A}$ [Ans.]

Comments: The value of I_{CEO} is always much greater than the value of I_{CBO} .

4.9.8 Input Characteristics

The input characteristic shows the relation between the input current, I_B , and the input voltage, V_{BE} , for a fixed value of collector-emitter voltage, V_{CE} . To draw the input characteristic curves, the $B-E$ junction is forward biased by a variable voltage source, V_{BB} , and the $B-C$ junction is reverse biased by another variable voltage source, V_{CC} , as shown in Fig.4-21. [for the same reason described for CBC, no series resistor is used.] Now, the value of V_{BE} is gradually adjusted and the value of I_B is measure for a constant value of V_{CE} using the variable source, V_{BB} . By plotting the data on a $V_{BE} - I_B$ coordinate system, an input characteristic curve is found. The process is repeated for different values of V_{CE} . All the graphs are plotted in the same coordinate system. This produces a **family of input characteristic curves** for CE configuration. A typical family of input characteristic curves for an NPN transistor is shown in Fig.4-22. Here also, the input is across the forward biased base-emitter junction, and hence the graph of input current (I_B) versus input

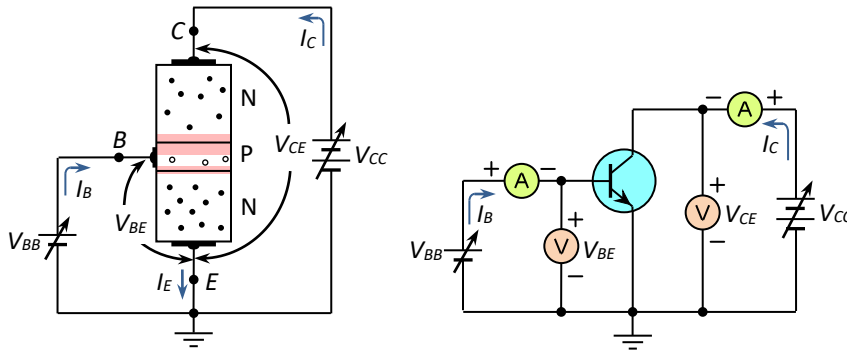


Fig.4-21: Circuit for determining transistor characteristics in CEC

voltage (V_{BE}) has exactly the same shape as that of a forward biased PN junction. The shape of the input characteristic curves resembles that of the CB configuration. The only difference is, input current, I_B , decreases with the increase of output voltage, V_{CE} . That's why the input curves shift to right direction with the increase of V_{CE} .

A larger value of V_{CE} results in larger reverse bias of the B - C junction which makes the depletion region wider and makes the base smaller. When the base is smaller, there are fewer recombinations of injected minority carriers. This results in a reduction in the base current. In contrast to the input characteristics of CB, here the input current is plotted in μA units as I_B is much smaller than I_C and I_E . The CE input characteristics are often called the **base characteristics**.

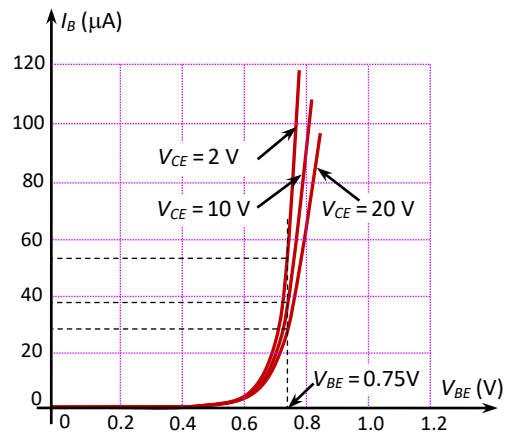


Fig.4-22: Common-emitter input characteristic (NPN)

4.9.9 Output Characteristics

The output characteristics can be determined using the same circuit of Fig.4-21. For determining this characteristics, the output voltage, V_{CE} , is gradually changed and the output current, I_C , is measured for different fixed values of input current, I_B . When the measured data are plotted on an I_C versus V_{CE} coordinate system, we get a family of output characteristic curves. These characteristics are also called **collector characteristics**. Fig.4-23 shows a typical family of output characteristic curves for an NPN transistor in CE configuration.

Here, also the graph has three regions. The region where I_C increases with V_{CE} is called the **saturation region**. The region below the $I_B = 0$ line is called the **cutoff region**. And the area in between the saturation region and cutoff region, where the curves are more or less parallel to the x -axis, is called the **active region**.

In the active region of CBC output characteristics, for a fixed value of I_E , the value of I_C (and hence the value of α) increases by a very small amount with the increase of output voltage, V_{CB} , that may not be detectable in the graphs. But in case of CEC the value of I_C (and hence the value of β) increases by a comparatively larger amount with the increase in output voltage, V_{CE} . It may be easily noticed from the graphs. For example, look at the points P and Q on the graph for $I_B = 40 \mu\text{A}$. For point P , $V_{CE} = 5 \text{ V}$ and $I_C = 3.8 \text{ mA}$. So the value of $\beta = (I_C \div I_B) = (3.8 \text{ mA} \div 40 \mu\text{A}) = 95$. On the other hand, for point Q , $V_{CE} = 10 \text{ V}$ and $I_C = 4 \text{ mA}$. So the value of $\beta = (4 \text{ mA} \div 40 \mu\text{A}) \approx 100$. That is, for the increase in V_{CE} from 5 V to 10 V, β has increased from 95 to 100. Moreover, for higher values of I_B , the rate of increase is even higher. What is the reason of this increase?

This increase simply reflects the fact that we have already discussed: The greater the value of reverse biased voltage, V_{CE} , of B - C junction, the wider the depletion region, the smaller the base region. Consequently, (as described in transistor action) the percentage of base recombination and hence, the base current decreases (this effect is called **Early effect**). On the contrary, percentage of the collector current increases. But as shown in Fig.4-22, for each graph, the value of base current I_B is kept fixed. To do that, more carriers have to be injected by the emitter into base. So more carrier will travel through the collector that results in increase of collector current for a fixed value of I_B with the increase of V_{CE} .

Unlike the CBC characteristics, here the saturation region lies on the right-hand side of the y-axis within values of V_{CE} approximately 0 to 0.7. Note in Fig.4-24 that the total C - E voltage, V_{CE} , is the sum of the small forward biasing voltage, V_{BE} , and the reverse biasing voltage, V_{CB} . Thus, $V_{CE} = V_{CB} + V_{BE}$, or, for silicon transistor, $V_{CB} \approx V_{CE} - 0.7 \text{ V}$. Hence, if V_{CE} is reduced to 0.7 V, V_{CB} must become 0, and the B - C junction is no longer reverse biased. If V_{CE} is reduced further, V_{CB} will be negative, i.e., B - C junction will be forward biased and I_C starts to fall off. As in this region, both the B - E junction and the B - C junction are forward biased, this is called the saturation region. For $V_{CE} = 0.2 \text{ V}$, the value of V_{CB} will be -0.5 V ($V_{CB} \approx 0.2 \text{ V} - 0.7 \text{ V} = -0.5 \text{ V}$). That is B - C junction will be forward biased by 0.5 V. For this forward bias the collector current will be zero. For this reason, the saturation value of V_{CE} , designated as $V_{CE(sat)}$, ranges from 0.1 V to 0.4 V (not -ve values like CB configuration).

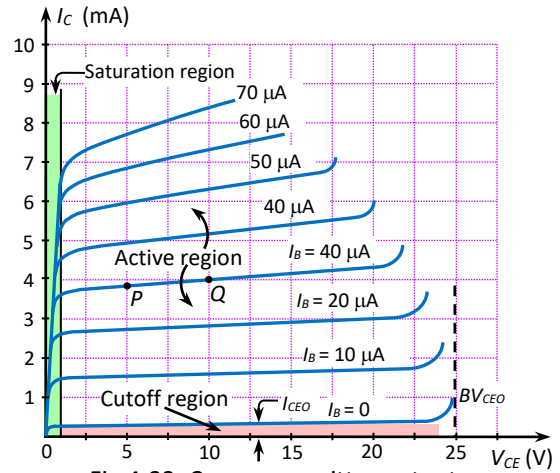


Fig.4-23: Common-emitter output characteristics (NPN)

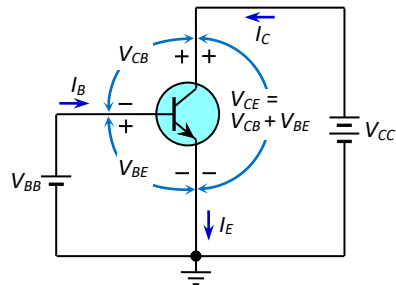


Fig.4-24: $V_{CE} = V_{CB} + V_{BE} \approx V_{CB} + 0.7 \text{ V}$

It is already mentioned that, the region below the $I_B = 0$ line is called the *cutoff region*. The cutoff collector current is denoted as I_{CEO} . Since, the value of I_{CEO} is much larger ($I_{CEO} \approx \beta I_{CBO}$), in CE configuration, the cutoff region is quite larger than that in CB configuration characteristics.

It is apparent in Fig.4-23 that the characteristic curves corresponding to larger values of I_B rise more rapidly to the right than those corresponding to smaller values of I_B . If these lines are projected to the left, as shown in Fig.4-25, it is found that they all intersect the x -axis at approximately the same point. This point of intersection is denoted as V_A and is called reverse **Early voltage** after J. M. Early, who first investigated this relation. Fig.4-25 shows an exaggerated view of the I - V characteristics plotted for constant values of (V_{BE}) base-emitter voltage, instead of base current, I_B . The curves are theoretically linear with respect to V_{CE} in the forward-active mode. The slope in these characteristics is due to an effect called **base-width modulation** that was first analyzed by J. M. Early. The phenomenon is generally called the **Early effect**. For a given value of V_{BE} (or I_B) in an NPN transistor, if V_{CE} increases, their reverse bias voltage on the C-B junction increases, which means that the width of the collector-depletion region increases. This in turn reduces the normal width of the base. Due to this reduced base width, the recombination of the majority carriers decreases and hence, the collector current increases. The typical values of V_A range from 50 V to 300 V. This same effect takes place in PNP transistor, as well.

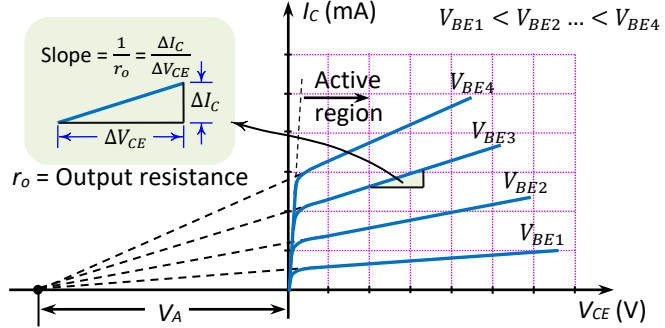


Fig.4-25: The reverse Early voltage, V_A (intersection on x -axis)

The linear dependence of I_C on V_{CE} in the forward active region, can be calculated as,

$$I_C = I_S(e^{V_{BE}/V_T}) \cdot \left(1 + \frac{V_{CE}}{V_A}\right) \quad (4-24)$$

The slope of an output characteristic curve depends on the Early Voltage. Again, we know that the output resistance is given by,

$$\frac{1}{r_o} = \frac{\Delta i_C}{\Delta v_{CE}} \approx \frac{di_C}{dv_{CE}} \quad (4-25)$$

Now using these two equations, Equ.(4-24) and Eu.(4-25) we can find,

$$\boxed{r_o \approx \frac{V_A}{I_C}} \quad (4-26)$$

where, I_C is the quiescent collector current when v_{BE} is a constant and v_{CE} is very small compared to V_A . In most cases, the dependence of i_C on v_{CE} is small and can be neglected in circuit design.

Example 4-7

The output resistance of a transistor is $r_o = 180 \text{ k}\Omega$ at $I_C = 1 \text{ mA}$. i) Determine the Early voltage at this current level. ii) Using this Early voltage, find r_o at $I_C = 0.1 \text{ mA}$ and iii) $I_C = 12 \text{ mA}$.

Solution:

i)

Using Equ.(4-), $V_A = r_o I_C = 180 \text{ k}\Omega \times 1 \text{ mA} = 180 \text{ V [Ans.]}$

ii)

Using Equ.(4-), $r_o \approx \frac{V_A}{I_C} = \frac{180 \text{ V}}{0.1 \text{ mA}} = 1.8 \text{ M}\Omega \text{ [Ans.]}$

iii)

Using Equ.(4-), $r_o \approx \frac{V_A}{I_C} = \frac{180 \text{ V}}{12 \text{ mA}} = 15 \text{ k}\Omega \text{ [Ans.]}$

Comments: From this example it is clear that the output resistance of a transistor in CE configuration, decreases with the increase in collector current (I_C).

Example 4-8

Calculate the output resistance in the active region of the output characteristics of a transistor in CE configuration shown in Fig. 4.24. for $I_B = 20 \text{ mA}$.

Solution:

Let us draw two vertical lines on $V_{CB} = 5 \text{ V}$ and $V_{CB} = 15 \text{ V}$ (as shown on the figure). Then, find the intersecting points of these lines with the $I_B = 20 \text{ mA}$ graph. Now, draw two horizontal lines through these points they will intersect the y-axis. These intersecting points give the values of I_C , which are shown on the graph.

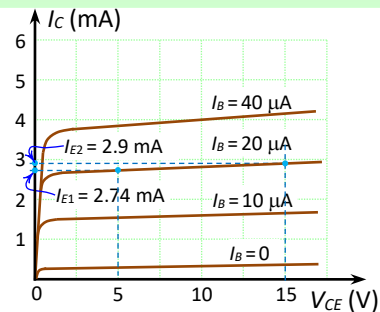


Fig.4-26: Graph for exercise number 4.

Now, $r_o = \frac{\Delta V_{CE}}{\Delta I_C} = \frac{15 \text{ V} - 5 \text{ V}}{2.9 \text{ mA} - 2.74 \text{ mA}} = 58 \text{ k}\Omega \text{ [Ans.]}$

Comments: I_C increases with the increase in base current (I_B). Output resistance decreases with the increase in collector current, I_C . For this reason, the slope of the output characteristic curve (in active region) increases with I_C .

4.9.10 Transfer Characteristics

The common-emitter transfer characteristics show the relations between the output current, I_C , and the input current, I_B , for a constant value of output voltage, V_{CE} . These are the graphs of I_C

versus I_B for different fixed values of V_{CE} . These are also called **current gain characteristics**. These characteristics can be derived from the output characteristic curves or can be measured.

To measure these graphs experimentally, the value of V_{CE} is kept constant at a suitable value and the value of input current, I_B , is adjusted step by step. At every steps of I_B , the value of I_C is measured. If the data are plotted on a $I_B - I_C$ coordinate system a transfer characteristic curve will be found. The process is repeated by fixing value of V_{CE} at different levels. In this way two or three curves are drawn. A set of typical transfer characteristic curves is shown in Fig.4-27. From this graph it is clear that there is a linear relationship between I_C and I_B (which depends on β). The slope of this curve gives the value of β . For the variation of output voltage, V_{CE} , there is a small change in the characteristic curves. The difference also increases with the increase in I_B .

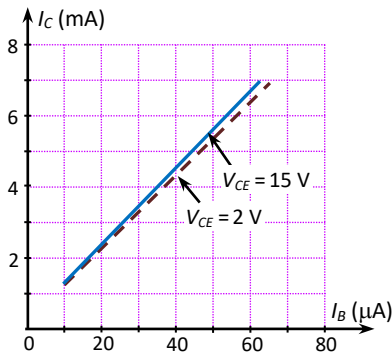


Fig.4-27: Transfer characteristic curves for CE configuration

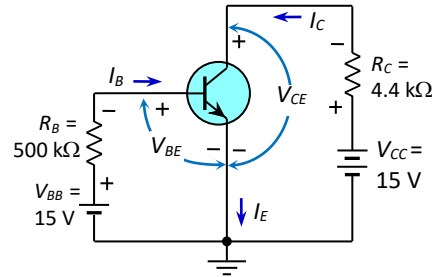


Fig. 4.28: Circuit for Example 4-9

Example 4-9

For the common-emitter circuit of Fig.4-28, determine I_B , I_C , I_E and V_{CE} . Assume a Si transistor with $\beta = 100$.

Solution:

In case of Si transistor, for the forward biased $B-E$ junction, we can consider

$$V_{BE} = 0.7 \text{ V}$$

Therefore,

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{15 \text{ V} - 0.7 \text{ V}}{500 \text{ k}\Omega} = 28.6 \text{ }\mu\text{A} \text{ [Ans.]}$$

Neglecting the leakage current (not given here), the collector current will be,

$$I_C = \beta I_B = 100 \times 28.6 \text{ }\mu\text{A} = 2.86 \text{ mA} \text{ [Ans.]}$$

$$I_E = I_B + I_C = 0.0286 \text{ mA} + 2.86 \text{ mA} = 2.8886 \text{ mA} \text{ [Ans.]}$$

Now, applying KVL in the output circuit we get,

$$V_{CC} - I_C R_C - V_{CE} = 0$$

or,

$$V_{CE} = V_{CC} - I_C R_C = 15 \text{ V} - (2.86 \text{ mA}) \times (4.4 \text{ k}\Omega) \approx 2.416 \text{ V} \text{ [Ans.]}$$

Comments: The actual value of V_{BE} may be little bit different from 0.7 V. But we approximate it to be $V_{BE} = 0.7$ V a reasonable consideration for Si transistors.

Example 4-10

Repeat Example 4-9, considering an ideal transistor, that is $V_{BE} = 0$ V.

Solution:

As $V_{BE} = 0$, the base current will be,

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{15 \text{ V} - 0 \text{ V}}{500 \text{ k}\Omega} = 30.0 \text{ }\mu\text{A} \text{ [Ans.]}$$

The collector current will be,

$$I_C = \beta I_B = 100 \times 30.0 \text{ }\mu\text{A} = 3.0 \text{ mA [Ans.]}$$

$$I_E = I_B + I_C = 0.03 \text{ mA} + 3.0 \text{ mA} = 3.03 \text{ mA [Ans.]}$$

The output voltage, V_{CE} , can be calculated using the same method in Example 4- as,

$$V_{CE} = V_{CC} - I_C R_C = 15 \text{ V} - (3.0 \text{ mA}) \times (4.4 \text{ k}\Omega) \approx 1.8 \text{ V [Ans.]}$$

Comments: As discussed in Chapter 2, in ideal case the forward voltage drop across a PN junction is 0. Similarly, here for an ideal transistor, V_{BE} has been considered as 0.

4.10 Characteristics of a Transistor in CCC

The third and final way to connect a transistor in a circuit is the *common-collector configuration* (CCC). In this configuration the collector terminal is made common to both the input circuit and the output circuit. That is, base and collector constitute the input circuit, and emitter and collector constitute the output circuit. Fig.4-29 shows the biasing process of both NPN and PNP transistor in CCC. V_{BB} reverse biases the $B-C$ junction, while V_{CC} forward biases the $B-E$ junction. Note that in this configuration, the input circuit includes the reverse biased junction and the output circuit includes the forward biased junction. Hence, only this configuration has very high input resistance

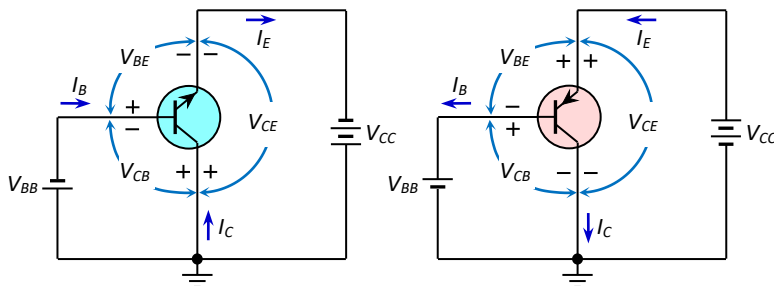


Fig.4-29: The common-collector bias configuration

and very low output resistance. For this reason, this CC configuration is used for impedance matching. In Fig.4-29, it is clear that

$$V_{CE} = V_{CB} + V_{BE} \quad (4-27)$$

or,

$$V_{CB} = V_{CE} - V_{BE} \quad (4-28)$$

In these circuits, as there is no resistors, connected between the battery and the transistor terminals, $V_{CE} = V_{CC}$ and $V_{CB} = V_{BB}$. V_{BE} is practically constant and for Si transistor its value is about 0.7 V (voltage drop of a forward biased PN junction). Thus,

$$V_{CB} = V_{BB} = V_{CC} - 0.7 \text{ V}$$

Therefore, in order to keep the B-C junction reverse biased ($V_{CB} > 0 \text{ V}$), V_{BB} must be larger than ($V_{CC} - 0.7 \text{ V}$). In this CC configuration, base-collector voltage, V_{CB} , is the input voltage and base current, I_B , is input current. On the other hand, emitter-collector voltage, V_{CE} , is the output voltage and emitter current, I_E , is the output current.

4.10.1 Input Characteristics

Now, let us study the input characteristics in this configuration. Certainly, the input characteristics will show the variation of input current, I_B , for the variation of input voltage, V_{CB} , with various fixed values of output voltage, V_{CE} . A set of input characteristic curves for an NPN transistor is shown in Fig.4-30. Look at the shape of the input characteristic curves. Of course, these are not the characteristics of a forward biased PN junction as found for CB and CE configurations. The graphs show that the base current, I_B , falls to zero rapidly as V_{CB} increases slightly. This behavior can be explained by the fact that V_{BE} must remain close to about 0.5 V to 0.7 V in order for any appreciable base current to flow. But, from Equ.(4-28),

$$V_{BE} = V_{CE} - V_{CB} \quad (4-29)$$

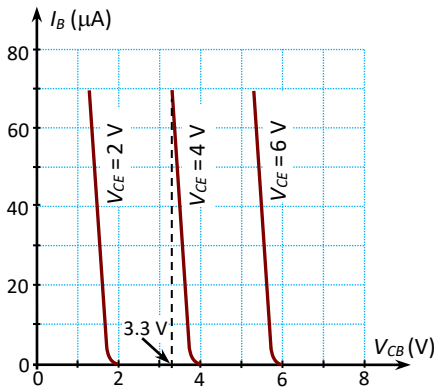


Fig.4-30: Common-collector input characteristic (NPN)

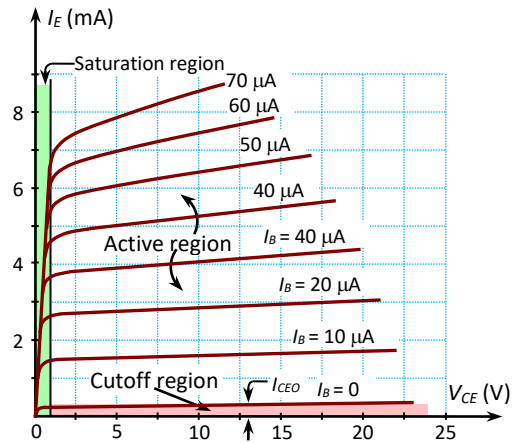


Fig.4-31: Common-collector output characteristic (NPN)

Therefore, if the value of V_{CB} is allowed to increase to a point where it is near the value of V_{CE} , the value of V_{BE} approaches zero. Thus, B-E junction loses its forward bias and base current falls to zero. Let us explain the situation a bit detailed, considering a graph in Fig.4-30 that corresponds to $V_{CE} = 4 \text{ V}$. When, $V_{CB} = 4.4$, from Equ.(4-29), we get $V_{BE} = 4 \text{ V} - 3.3 \text{ V} = 0.7 \text{ V}$. Then, the B-E

junction is forward biased and we get a substantial amount of base current ($I_B \approx 70\mu\text{A}$). If now V_{CB} is allowed to increase to 4 V, then $V_{BE} = 4\text{ V} - 4\text{ V} = 0\text{ V}$. So the B - E junction is no longer forward biased which results in $I_B = 0$. Note in the figure that $I_B = 0$, when $V_{CE} = V_{CB} = 4\text{ V}$.

Regarding the input characteristics of CC configuration one more point to remember is that the input current is highly dependent on the output voltage. I_B changes in the same order of the V_{CE} . That is, if V_{CE} changes from 4 V to 6 V, I_B will also shift from 4 V to 6 V on the x -axis.

4.10.2 Output Characteristics

The output characteristics show the relation between the output current, I_E , and the output voltage, V_{CE} , for different constant values of input current, I_B . A typical family of output characteristic curves for an NPN transistor in CC configuration is shown in Fig.4-31. Compare this graph with that of CE configuration. The only difference is the dependent parameter (parameter on the y -axis). For CC configuration it is I_E and for CE configuration it is I_C . But, we know that $I_E \approx I_C$ or I_E is greater than I_C by a very small amount (maximum 2%). For this reason, the output characteristics of CCC transistor has exactly the same pattern as the CEC transistor. Compare the graphs of Fig.4-23 and Fig.4-31, the only change is: I_E is slightly greater than I_C which is hardly noticeable. As discussed before, these characteristics also have three regions which are same as the CE configuration. If the base terminal is kept open $I_B = 0$, exactly the same leakage current flows here. This is shown in Fig.4-32.

4.10.3 Calculation of Leakage Current in CCC

We know that
$$I_E = I_C + I_B$$

And,
$$I_C = \alpha I_E + I_{CBO}$$

Putting the value of I_C into the first equation

we get,
$$I_E = \alpha I_E + I_{CBO} + I_B$$

or,
$$I_E(1 - \alpha) = I_B + I_{CBO}$$

Dividing both sides by $(1 - \alpha)$ we get,

or,
$$I_E = \frac{I_B}{(1 - \alpha)} + \frac{1}{(1 - \alpha)} I_{CBO} \quad (4-30)$$

When the base is open (Fig.4-32), $I_B = 0$. Therefore, the value of the leakage current in CC configuration is,

$$I_{E(\text{Leakage})} = \frac{1}{(1 - \alpha)} I_{CBO}$$

Using Equ.(4-18),
$$I_{E(\text{Leakage})} = I_{CEO}$$

From this equation, we find that the same leakage current flows in common collector configuration.

4.10.4 Transfer Characteristics

The common-collector transfer characteristics show the relations between the output current, I_E , and the input current, I_B , for a constant values of output voltage, V_{CE} . These are also called current gain characteristics. These characteristics can be derived from the output characteristic curves or

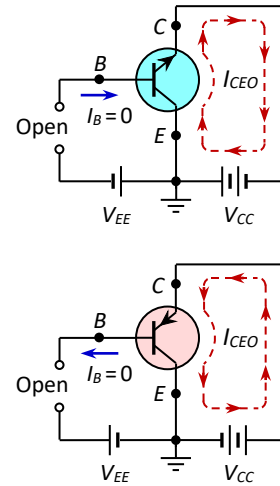


Fig.4-32: Leakage current in CC configuration

can be measured. To measure these graphs experimentally, the value of V_{CE} is kept constant at a suitable value and the value of input current, I_B , is adjusted step by step. At every steps of I_B , the value of I_E is measured. If the data are plotted on a $I_B - I_E$ coordinate system, a transfer characteristic curve will be found. The process is repeated by fixing value of V_{CE} at different levels. In this way, two or three curves are drawn. A set of typical transfer characteristic curves is shown in Fig.4-33. As the value of I_E is approximately equal to I_C , this graphs are also similar to that of the CE configuration.

4.10.5 Current Gain in CC Configuration

Among the three configurations, CC configuration gives the maximum current gain. It is defined as the ration of the output current, I_E , to the input current, I_B , and is denoted by γ . Mathematically

$$\gamma_{dc} = \frac{I_E}{I_B} \quad [\text{for DC signal}]$$

$$\gamma_{ac} = \frac{\Delta I_E}{\Delta I_B} \quad [\text{for AC signal}]$$

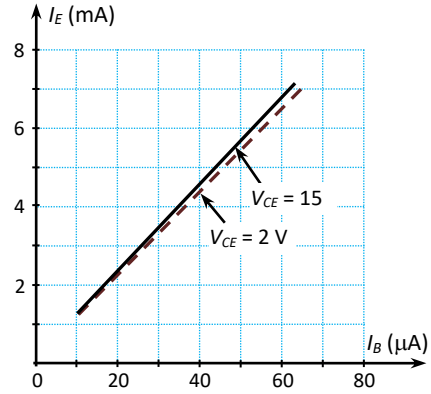


Fig.4-33: Transfer characteristic curves for CE configuration.

4.10.6 Relation among α , β , and γ

We know that for any configuration of any type of transistors,

$$I_E = I_C + I_B$$

Dividing both sides by I_B ,

$$\frac{I_E}{I_B} = \frac{I_C}{I_B} + 1$$

or,

$$\boxed{\gamma = \beta + 1} \quad (4-31)$$

or,

$$\gamma = \frac{\alpha}{1 - \alpha} + 1 = \frac{\alpha + 1 - \alpha}{1 - \alpha}$$

$\therefore$

$$\boxed{\gamma = \frac{1}{1 - \alpha}} \quad (4-32)$$

4.10.7 Exact Relation Between the Input Current and Output Current

We have already found that, $I_E = \frac{1}{(1 - \alpha)} I_B + \frac{1}{(1 - \alpha)} I_{CBO}$ From Equ.(4-30)

Using Equ.(4-30),

$$\boxed{I_E = \gamma I_B + \gamma I_{CBO}} \quad (4-33)$$

Example 4-11

The collector cutoff current of a Si transistor is 50 nA and $\alpha = 0.995$. If 50 μ A current flows in the base then determine (i) β and γ (ii) I_{CEO} (iii) I_E and I_C .

Solution:

(i)

Using Equ.(4-15),
$$\beta = \frac{\alpha}{1 - \alpha} = \frac{0.995}{1 - 0.995} = 199 \text{ [Ans.]}$$

Using Equ.(4-31),
$$\gamma = (\beta + 1) = (199 + 1) = 200 \text{ [Ans.]}$$

(ii)

Given collector cutoff current
$$I_{CBO} = 50 \text{ nA}$$

Using Equ.(4-19),
$$I_{CEO} = (\beta + 1)I_{CBO} = (199 + 1)(50 \text{ nA}) = 10 \text{ }\mu\text{A [Ans.]}$$

(iii)

Using Equ.(4-20),
$$I_C = \beta I_B + I_{CEO} = 199 \times (50 \text{ }\mu\text{A}) + 10 \text{ }\mu\text{A} = 9.96 \text{ mA [Ans.]}$$

Using Equ.(4-7),
$$I_C = \alpha I_E + I_{CBO}$$

or,
$$I_E = \frac{I_C - I_{CBO}}{\alpha} = \frac{9.96 \text{ mA} - 50 \text{ nA}}{0.995} = 10 \text{ mA [Ans.]}$$

Comments: The difference between β and γ is only 1, so in most of literature γ is not considered.

Example 4-12

Common collector circuit can be represented as shown in Fig. 4.42. For the component values shown in the figure, calculate: (i) I_B , I_C , and I_E (ii) V_{CE} , V_{CB} , V_E , and V_B . Assume $V_{BE} = 0.7 \text{ V}$ and $\beta = 75$.

Solution:

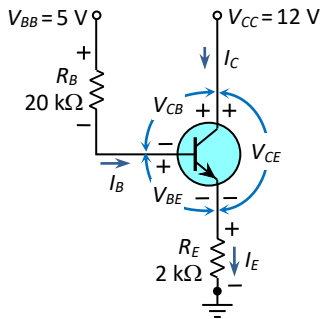


Fig.4-34: Circuit for Example 12

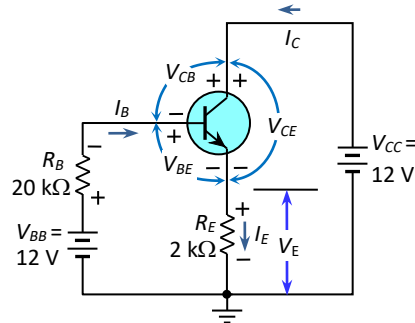


Fig.4-35: Fig. 4.42 redrawn

To understand the circuit properly, we can redraw it as shown in Fig.4-42.

(i)

Applying KVL in the base-emitter circuit, we get,

$$V_{BB} - I_B R_B - V_{BE} - V_E = 0$$

or,
$$V_{BB} - I_B R_B - V_{BE} - I_E R_E = 0$$

or,
$$V_{BB} - I_B R_B - V_{BE} - (\beta + 1)I_B R_E = 0 \quad [\because I_E = (\beta + 1)I_B]$$

or,

$$I_B = \frac{V_{BB} - V_{BE}}{R_B + (\beta + 1)R_E}$$

$\therefore$

$$I_B = \frac{5 \text{ V} - 0.7}{20 \text{ k}\Omega + (75 + 1) \times (2 \text{ k}\Omega)} = 25 \mu\text{A} \text{ [Ans.]}$$

Using Equ.(4-13),

$$I_C = \beta I_B = 75 \times 25 \mu\text{A} = 1.875 \text{ mA} \text{ [Ans.]}$$

Using Equ. (4-8),

$$I_E = I_C + I_B = 1.875 \text{ mA} + 25 \mu\text{A} = 1.9 \text{ mA} \text{ [Ans.]}$$

(ii)

Now, applying KVL in the collector-emitter circuit, we get,

$$V_{CC} - V_{CE} - V_E = 0$$

or ,

$$V_{CE} = V_{CC} - I_E R_E [\because V_E = I_E R_E]$$

$\therefore$

$$V_{CE} = 12 \text{ V} - (1.9 \text{ mA}) \times (2 \text{ k}\Omega) = 8.2 \text{ V} \text{ [Ans.]}$$

From the circuit, we can write,

$$V_{CE} = V_{CB} + V_{BE}$$

or

$$V_{CB} = V_{CE} - V_{BE} = 8.2 \text{ V} - 0.7 \text{ V} = 7.5 \text{ V} \text{ [Ans.]}$$

Again,

$$V_E = I_E R_E = (1.9 \text{ mA}) \times (2 \text{ k}\Omega) = 4.8 \text{ V} \text{ [Ans.]}$$

And,

$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 4.8 \text{ V} = 4.5 \text{ V} \text{ [Ans.]}$$

Comments: DC analysis of a CCC circuit is exactly same as that of CEC. Its AC analysis will be different that will be discussed in **Chapter Habib**.

4.11 Maximum Transistor Ratings

A transistor, like any other electronic device, has some limitations on its operation. These limitations are stated in the form of maximum ratings and are normally specified on the manufacturer's data sheet. Typically, maximum ratings are given for collector-to-base voltage (V_{CBO}), collector-to-emitter voltage (V_{CEO}), emitter-to-base voltage (V_{BEO}), collector current (I_{CM}), and total power dissipation (P_{tot}) which is sometimes represented as Maximum Power Dissipation ($P_{D(max)}$). The product of operating V_{CE} and I_C must not exceed the maximum power dissipation. Both V_{CE} and I_C cannot be maximum at the same time. For a given value of V_{CE} the value of the maximum I_C can be

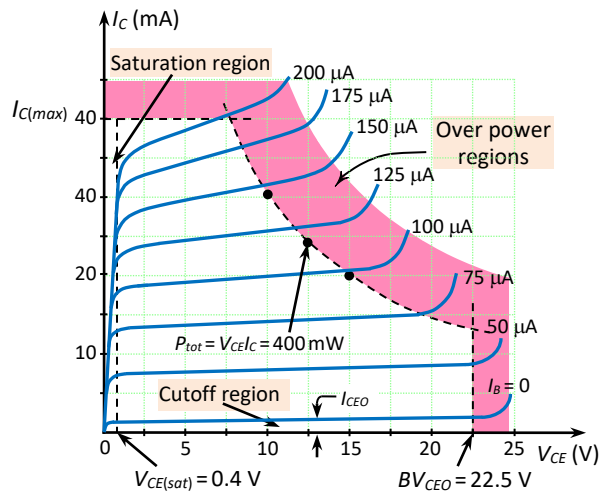


Fig.4-36: Maximum power dissipation curve

calculated as

$$I_C = \frac{P_{tot}}{V_{CE}} \quad (4-34)$$

Similarly, if the value of I_C is given, the maximum emitter-emitter voltage can be calculated as,

or,

$$V_{CE} = \frac{P_{tot}}{I_C} \quad (4-35)$$

For any given transistor, a maximum power dissipation curve can be plotted on the collector characteristic using Equ.(4-34). Considering a transistor of $P_{tot} = 400$ mW, its maximum power line has been draw in this figure (dash line). For the safe and linear operation of a transistor, we have to avoid the saturation region, cutoff region and also the over-power regions.

4.11.1 Derating of $P_{D(max)}$ or P_{tot}

In the manufacturers' datasheets, P_{tot} (or $P_{D(max)}$) is usually specified at 25°C. For higher temperatures, the value of P_{tot} decreases. Datasheets often give derating factors for determining P_{tot} at any temperature above 25°C. For example, a derating factor of 2 mW/°C indicates that the maximum power dissipation is reduced 2 mW for each degree Celsius increase in temperature above 25°C.

Example 4-13

Suppose a transistor has $P_{tot} = 2$ W at 25°C. The derating factor for P_{tot} of this transistor is 10 mW/°C. Determine the value of P_{tot} of this transistor at $T_j = 75^\circ\text{C}$.

Solution:

The change (decrease) in P_{tot} can be calculated as,

$$\Delta P_{tot} = (10 \text{ mW})(75^\circ\text{C} - 25^\circ\text{C}) = 0.5 \text{ W}$$

Therefore, the deteriorated maximum power of the transistor at 75°C will be,

$$P_{tot(75^\circ)} = 2 \text{ W} - 0.5 \text{ W} = 1.5 \text{ W [Ans.]}$$

Conclusion: If the operating temperature of a transistor is not controlled properly, the maxium power capacity may fall drastically, which may cause catastrophic damage of the device.

4.12 Power Transistor and Heat Sinks

Depending of the power handing capacity, there are two types of transistors. One is **small-signal transistor** and the other is **power transistor** (or **large signal transistor**) The current and voltage levels handldged by a small-signal transistor is very small compared to the supply voltage of the circuit. These transistors are physically small and are manufactured in plastic (thermoset or thermoplastic), or ceramic. As shown in Fig.4-40, the most common package types of small-signal transistors are- TO-92, TO-254. The typical power capacity of these transistors is less than 1 Watt. As the base layer is very simll, these transistors have very larger values of β . On the other hand, a power transistor can hangle larger voltage and current. Though, power transistors are also

packaged in plastic/ceramic cover, they have a metal part (internally connected to collector). The physical size of power transistors is comparatively larger. As the base layer is large, the values of β of these transistors are comparatively low. The common types of power transistor package are: TO-247, TO-220, TO-4, TO-66 as shown in Fig.4-40. These transistors can handle from 1 Watt to few 100 Watt.

One way to increase the power handling capacity of a transistor is to remove internal heat faster, using heat-sink. **Heat sink** is a bulk of metal (mainly aluminum) that has larger surface area to dissipate heat to air. The heat-sink may have many fins or tabs as shown in Fig.4-37. The heat-sinks are attached to the metal part of the transistor sometimes directly or sometimes using an insulating sheet called mica. Mica is an insulating material but a good conductor of heat.

No matter what kind of heat sink is used, its purpose is to lower the case temperature because this will lower the internal or junction temperature of the transistor. The data sheet includes other quantities called thermal resistances. These allow a designer to work out the case temperature for different heat sinks. In Chapter (**Power Amplifier**) heat-sinks are discussed more elaborately.

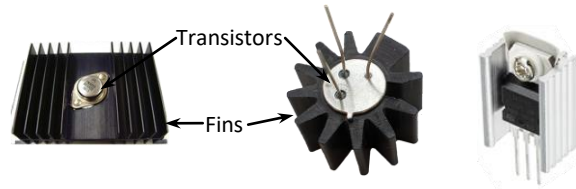


Fig.4-37: Heat sink of different shapes and sizes

4.13 BJT In Digital Logic Circuits

We have already discussed about Logic Gates (building blocks of digital circuits) in Section 3-6. There are three types of fundamental logic gates. In Section 3-6, we have implemented AND gate and OR gate using diodes and resistors. The third fundamental logic gate is the **NOT gate**. It is also called the **inverter**. This inverter can be implemented using transistors, but not using diodes. In fact, many other logic gates can be constructed using transistors.

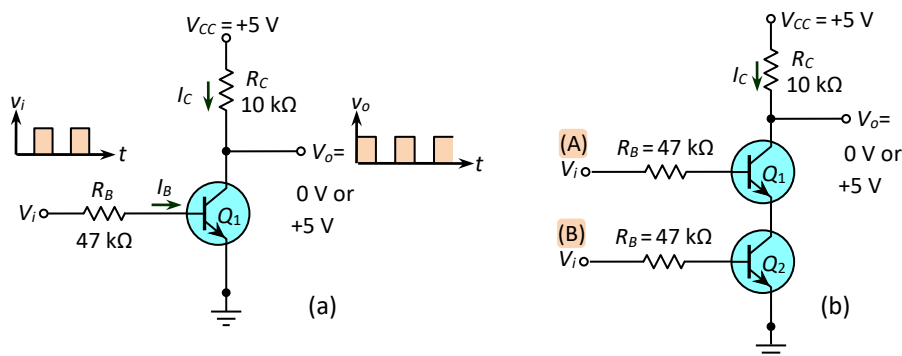


Fig.4-38: Implementation of Logic Gates using BJT: (a) Inverter or NOT Gate, and (b) NAND Gate

The logic gates shown in Fig.4-38, are older types of circuit technology called **RTL (Resistor-Transistor Logic)** used to illustrate the basic idea of how transistors can be used in switching

applications. The practically used technique for implementing logic gates is called **TTL (Transistor-Transistor Logic)**, which is normally studied in Digital Electronics Books.

The inverter (NOT gate) changes voltage at one level to the opposite level. When a HIGH-level voltage (logically '1') is applied to the input of the inverter, shown in Fig.4-38(a), a LOW-level voltage (logically '0') appears at the output and vice versa.

The **NAND gate** produces a LOW-level output voltage, when and only when all of its inputs are at a HIGH-level voltage; otherwise the output is a HIGH-level. A simple two-input NAND gate is shown in Fig.4-38(b). When high-level voltages are applied to the inputs of transistors Q_1 and Q_2 , both transistors are ON (saturated) and a LOW-level voltage is developed at the collector of Q_1 because of the direct connection to ground, through the saturated (ON) transistors. When one or both of the inputs are at a LOW-level, the output is at a HIGH level, as the transistor(s) works as open switch.

The **NOR gate** produces a HIGH-level output voltage when and only when all of its inputs are at a LOW-level voltage; otherwise the output is a HIGH-level. A simple two-input NOR gate is shown in Fig.4-38(c). When HIGH-level voltages are applied to any input of the transistors Q_1 and Q_2 , they will be saturated (fully ON), and output at the collector of this transistor will be LOW (or '0'). As the transistors are connected in parallel, any ON transistor will produce LOW voltage. When both of the inputs are at a LOW-level, the transistors will be OFF and their collector voltage will be HIGH and hence, the output will also be HIGH.

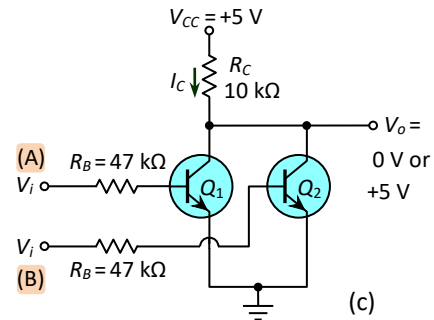
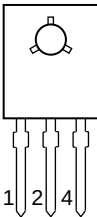


Fig. 4.38: Implementation of Logic Gates using BJT: (c) NOR Gate

4.14 Transistor Specifications

In modern electronic circuits, discrete transistors are used primarily for applications in which only one or a small number of devices are required, and in applications where substantial power is dissipated. In many applications, both discrete and integrated components are used. In these applications, the integrated circuit typically performs complex, low-power **signal conditioning**, and discrete transistors then drive power-consuming loads such as indicator lamps, motors or loud speakers.

To select a transistor for a particular application, we must study its **specification sheet** (or **datasheet**). To get some idea, the specification sheet of BC145 (a very common transistor in electronics laboratory) is given in Fig.4-39. The maximum ratings (eg., V_{CE0} , I_{CM} etc.) in this figure, show the maximum voltage between two terminals and the maximum current through a terminal that the transistor can sustain. The maximum power dissipation of the transistor is $P_{tot} = 8$ Watt at ambient temperature of 25°C . For higher temperature the transistor can tolerate less power than P_{tot} rating. Except f_T all of the parameters are self-explanatory. f_T is the **unity current gain cutoff frequency**. It will be explained in details in high-frequency transistor modeling (Chapter 6). The graph of Fig.4-39 shows the variation of β with the collector current I_C . In addition to this graph every specification sheet includes many other electrical characteristics, which are not shown here.

Philips Semiconductors	Product specification								
NPN power transistor	BD145; BD147; BD149								
FEATURES • High current (max. 1.5 A) • Low voltage (max. 80 V)	PINNING <table><tr><th>PIN</th><th>DESCRIPTION</th></tr><tr><td>1</td><td>emitter</td></tr><tr><td>2</td><td>collector, connected to metal part of mounting surface</td></tr><tr><td>3</td><td>base</td></tr></table>	PIN	DESCRIPTION	1	emitter	2	collector, connected to metal part of mounting surface	3	base
PIN	DESCRIPTION								
1	emitter								
2	collector, connected to metal part of mounting surface								
3	base								
APPLICATIONS • Driver stages in hi-fi amplifiers and television circuits.									
DESCRIPTION NPN power transistor in a TO-126; SOT42 plastic package. PNP components: BD146, DB148 and BD 140.	 Fig.1 Simplified outline (TO-126; SOT42) and symbol.								

LIMITING VALUES

In accordance with the Absolute Maximum Rating System (IEC 144).

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
V_{CBO}	collector-base voltage	open emitter			
	BD145		-	45	V
	BD147		-	60	V
	BD149		-	100	V
V_{CEO}	collector-emitter voltage	open base			
	BD145		-	45	V
	BD147		-	60	V
	BD149		-	80	V
V_{EBO}	emitter-base voltage	open collector	-	5	V
I_C	collector current (DC)		-	1.5	A
I_{CM}	peak collector current		-	2	A
I_{BM}	peak base current		-	1	A
P_{tot}	total power dissipation	$T_{mb} \leq 70^\circ\text{C}$	-	8	W
T_{stg}	storage temperature		-65	+150	$^\circ\text{C}$
T_j	junction temperature		-	150	$^\circ\text{C}$
T_{amb}	operating ambient temperature		-65	+150	$^\circ\text{C}$

THERMAL CHARACTERISTICS

SYMBOL	PARAMETER	CONDITIONS	VALUE	UNIT
$R_{th\ j-a}$	thermal resistance from junction to ambient	note 1	100	K/W
$R_{th\ j-mb}$	thermal resistance from junction to mounting base		10	K/W

Note

1. Refer to TO-126; SOT42 standard mounting conditions.

CHARACTERISTICS

$T_j = 25\ ^\circ\text{C}$ unless otherwise specified.

SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
I_{CBO}	collector cut-off current	$I_E = 0; V_{CB} = 40\text{ V}$	-	-	100	nA
		$I_E = 0; V_{CB} = 40\text{ V}; T_j = 125\ ^\circ\text{C}$	-	-	10	μA
I_{EBO}	emitter cut-off current	$I_C = 0; V_{EB} = 5\text{ V}$	-	-	100	nA
h_{FE}	DC current gain	$V_{CE} = 2\text{ V};$ $I_C = 5\text{ mA}$	40	-	-	
		$I_C = 150\text{ mA}$	64	-	250	
		$I_C = 500\text{ mA}$	25	-	-	
		$I_C = 150\text{ mA}; V_{CE} = 2\text{ V}$	64	-	160	
V_{CEsat}	C-E saturation voltage	$I_C = 500\text{ mA}; I_B = 50\text{ mA}$	-	-	0.5	V
V_{BE}	base-emitter voltage	$I_C = 500\text{ mA}; V_{CE} = 2\text{ V}$	-	-	1	V
f_T	transition frequency	$I_C = 50\text{ mA}; V_{CE} = 5\text{ V};$ $F = 100\text{ MHz}$	-	190	-	MHz
$\frac{h_{FE1}}{h_{FE2}}$	DC current gain ratio of the complementary pairs	$ I_C = 150\text{ mA}; V_{CE} = 2\text{ V}$	-	1.4	1.6	

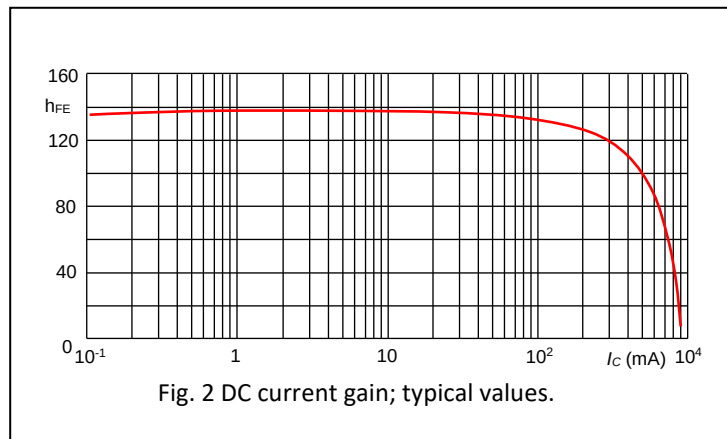


Fig.4-39: The data sheet of a common transistor BC145 (Courtesy of Philips Semiconductor Con.)

4.15 Packaging of Transistors

Discrete transistors are packaged in a wide variety of metal and plastic enclosures (cases). Fig.4-40 shows some of the common case types. Which are identified by **TO-numbers** (**Transistor Outline**)

and **SOT-numbers** (**Small Transistor Outline**). Almost all of the low-power (small-signal) transistors come with plastic cases (e.g., TO-92 and TO-254), while medium and high-power transistors come in plastic package with metal collector or full metal package (e.g., TO-218, TO-220, TO-247 and TO-4, TO-66). In some power transistors, the collector is attached to the metal case to dissipate high temperature (e.g., TO-247, TO-264, TO-4, TO-66). All manufacturers use the same leads pattern for the same TO-number.



Fig.4-40: Photographs of some transistors with their packaging number

For example the position of base, emitter and collector for some TO-number are shown in Fig. 4.41. Some TO-number has different variants, so their pin arrangements are different.

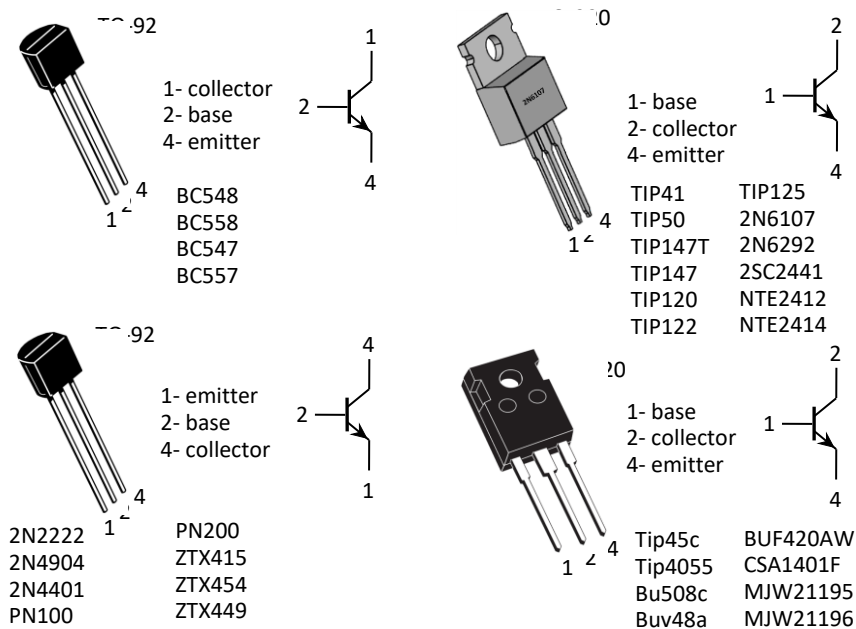


Fig.4-41: Pin diagrams of some TO packages

4.16 Transistor Testing

In many situations, we have to check a transistor, whether it is functional or not. For example, if a electronic system, built using discrete components, does not work, we may have to check the transistors of this systems. Moreover, in Analog Electronics Laboratory, a transistor may be functional, because it is used repeatedly by the students. In that case, before connecting the transistor in the experimental circuit, we should test the transistor. There are two ways to test a transistor: Off-circuit testing and On-circuit testing as described below.

4.16.1 Off-circuit Tests

Off-current tests mean, to check a transistor when not connected in a circuit board. This can be done by using a digital multimeter very easily. Generally, a digital multimeter has a facility to measure the β (h_{FE}). As shown in Fig.4-42, the multimeter has eight connecting holes for this purpose. Four holes for NPN transistor, and four holes for PNP transistor. If we know the type and three terminals (B, E, C) of the transistor, we have to insert it into the appropriate holes, i.e., the B, E, C terminals of the transistor into the B, E, C holes of the multimeter. The multimeter must show the value of β (h_{FE}) of the transistor if it is OK. If the multimeter does not show the value of β (h_{FE}) the transistor is not operational. If we know the type of the transistor but don't know the B, E, C terminal, then we have to repeat the testing **four times** taking three holes from four. But, if we even do not know the type and terminals of the transistor, first we have to find the type of the transistor.

To find the type of a transistor, set the multimeter in $k\Omega$ range. By connecting the +ve probe and the -ve probe of the multimeter to different leads of the transistor, find out the common lead that gives low resistance reading with the other two leads. This common lead is the base of the transistor. If the +ve probe of the meter, connected to the common terminal gives low reading the transistor is an NPN transistor. On the other hand, if the -ve probe of the meter, connected to the common lead, gives low resistance reading the transistor is PNP transistor. In this way, the type of the transistor and the base of the transistor can be determined. Now, we can measure the β (h_{FE}) and find out the other two terminals. If the multimeter does not have the β (h_{FE}) measuring facility, or even the meter is an analog meter, it is also possible to determine the type and three terminals of the transistor. Due to the complicated description, that process is not given here.



Fig.4-42: Measurement of h_{FE} of a transistor using multimeter

4.16.2 On-circuit Tests

In this process, the transistor is tested keeping it connected in the circuit board and with the power supplied to the circuit, ON. It is a very quick method to determine whether the transistor is working or not. The problem of this method is you must know the B, E, C terminals and the type of the transistor. As shown in Fig.4-43, a voltmeter is connected across the collector terminal and ground. The voltmeter should give some voltage reading which is less than V_{CC} . Now temporarily short the base and emitter terminals of the transistor, the meter reading should jump to V_{CC} for NPN transistor and the meter reading should be zero for PNP transistor. In this situation, we can say that the transistor is operational. But for NPN, if the meter reading is V_{CC} without shorting the base and emitter, the transistor may be open. Or if the meter reading does not jump to V_{CC} when the base-emitter is short, the collector-emitter of the transistor is short. Similarly for PNP transistor, if the meter reading is zero without shorting, then the transistor is open.

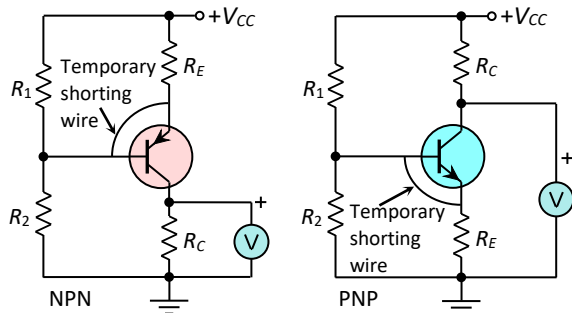


Fig.4-43: On-circuit test of transistor

4.17 Phototransistor

A phototransistor is similar to a regular BJT (bipolar junction transistor) except that the base current is produced and controlled by light instead of a base voltage. The phototransistor effectively converts light energy to an electrical signal. In a phototransistor the base current is produced when light falls on a photosensitive semiconductor base region. The collector-base PN junction is exposed to light through an opening (sometimes may have a lens) in the transistor package [see Fig.4-44(c)]. When there is no incident light, there is only a small thermally generated

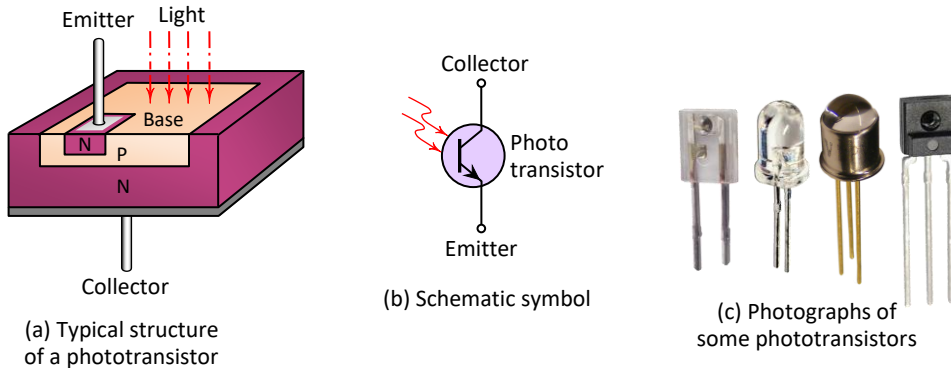


Fig.4-44: Phototransistor

collector-to-emitter leakage current, I_{CEO} which is called the **dark current**. Its typical range is few nA. When light strikes the collector-base PN junction, the thermally generated minority carrier concentration increases and a base current, I_p , is produced that is directly proportional to the light intensity. This action produces a collector current that increases with I_p . Except for the way base current is generated, the phototransistor behaves as a conventional BJT. In many cases, there is no electrical connection to the base. The relationship between the collector current and the light-generated base current in a phototransistor is,

$$I_C = \beta_{dc} I_p \quad (4-36)$$

The structural blocks, schematic symbol, and photographs of some typical phototransistors are shown in Fig.4-44. Since the actual photo-generation of base current occurs in the collector-base

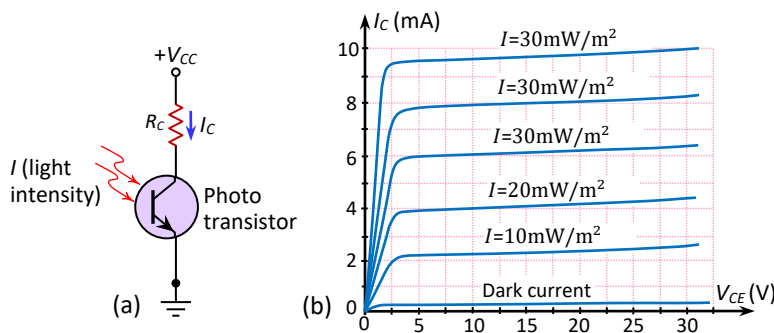


Fig.4-45: (a) Phototransistor circuit, (b) Typical I - V characteristics

region, the larger the physical area of this region, the more base current is generated. Thus, a typical phototransistor is designed to offer a large area to the incident light, as the simplified structure diagram in Fig.4-44(a) illustrates.

A phototransistor can be either a two-lead or a three-lead device. In the three-lead configuration, the base lead is brought out so that the device can be used as a conventional BJT with or without the additional light-sensitivity feature. In the two-lead configuration, the base is not electrically available, and the device can be used only with light as the input. In many applications, the phototransistor is used in the two-lead version.

Fig.4-45 shows a phototransistor with a biasing circuit, and the typical collector characteristic curves. Notice that each individual curve on the graph corresponds to a certain value of light intensity (in this case, the units are mW/cm^2) and that the collector current increases with light intensity. Phototransistors are not sensitive to all light but only to light within a certain range of wavelengths. They are most sensitive to particular wavelengths in the red and infrared part of the spectrum. The spectral response of a typical phototransistor is shown in Fig.4-46. The peak response is found for the 700 nm wavelength or near infrared.

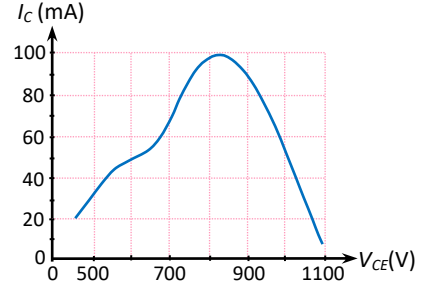


Fig.4-46: Typical phototransistor spectral response

Example 4-14

The phototransistor used in the circuit of Fig.4-47, operates in its linear range. For a light input of $1 \text{ lm}/\text{m}^2$, the photo current generated in the base is $0.1 \mu\text{A}$. Calculate the range of input light intensity that can be measured using this circuit. Assume that the saturation voltage of the photo transistor is $V_{CE(\text{sat})} = 1.2 \text{ V}$, $\beta_{DC} = 250$, and its collector dark (leakage) current is zero.

Solution:

Using KVL, $V_{CC} - I_C R_C - V_{CE}$

$$\text{or, } I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

Maximum collector current will flow, when the transistor saturates. Therefore,

$$I_{C(\text{max})} = \frac{V_{CC} - V_{CE(\text{sat})}}{R_C} = \frac{16 \text{ V} - 1.2 \text{ V}}{2.2 \text{ k}\Omega} \approx 6.73 \text{ mA}$$

Maximum base (photo) current will be,

$$\text{Using Equ.(4-34), } I_{p(\text{max})} = \frac{I_{C(\text{max})}}{\beta_{dc}} = \frac{6.73 \text{ mA}}{250} \approx 26.92 \mu\text{A}$$

So, the maximum light intensity will be,

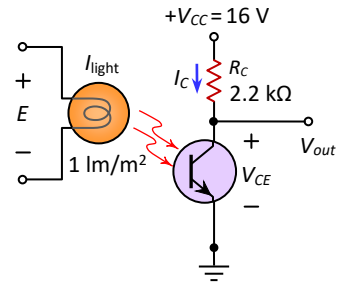


Fig. 4.47: Phototransistor circuit for Example 14

$$I_{\text{light(max)}} = \frac{26.92 \mu\text{A}}{0.1 \mu\text{A}} \times 1 \frac{\text{lm}}{\text{m}^2} \approx 270 \text{ lm/m}^2$$

Comments: The operating range (max lm/m²) of this light detecting circuit can be increased by decreasing the value of R_C , but the sensitivity will decrease consequently.

4.17.1 Applications

Phototransistors are used in a variety of applications. A light-operated relay circuit is shown in Fig.4-48(a). The phototransistor Q_1 drives the BJT Q_2 . When there is sufficient incident light on Q_1 , transistor Q_2 is driven into saturation, and collector current through the relay coil energizes it. The diode across the relay coil prevents, by its limiting action, a large voltage transient from occurring at the collector of Q_2 when the transistor turns OFF.

Fig.4-48(b) shows a circuit in which a relay is deactivated by incident light on the phototransistor. When there is insufficient light, transistor Q_2 is biased ON, keeping the relay energized. When there is sufficient light, phototransistor Q_1 turns ON; this pulls the base of Q_2 low, thus turning Q_2 OFF and de-energize the relay.

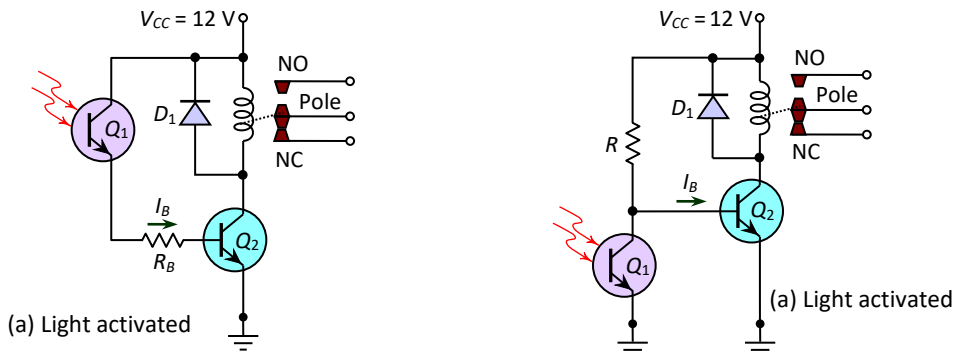


Fig.4-48: Relay controlling circuit using light

4.18 Optocouplers

An optocoupler uses an LED optically coupled to a photodiode or a phototransistor in a single package. Two basic types are: LED-to-photodiode and LED-to-phototransistor, as shown in Fig.4-49. Examples of typical packages are shown in Fig.4-49(c).

A key parameter in optocouplers is the **CTR (Current Transfer Ratio)**. The CTR is an indication of how efficiently a signal is coupled from input to output and is expressed as the ratio of a change in the LED current to the corresponding change in the photodiode or phototransistor current. It is usually expressed as a percentage.

$$\text{CTR} = \frac{I_{\text{out}}}{I_{\text{in}}} \text{ or } = \frac{\Delta i_{\text{out}}}{\Delta i_{\text{in}}} \quad (4-37)$$

Fig.4-50 shows a typical graph of CTR versus forward LED current. For this case, it varies from about 50% to about 110%.

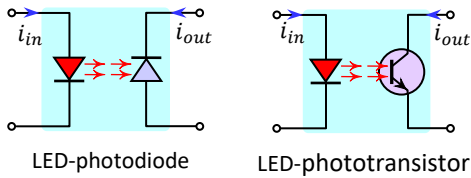
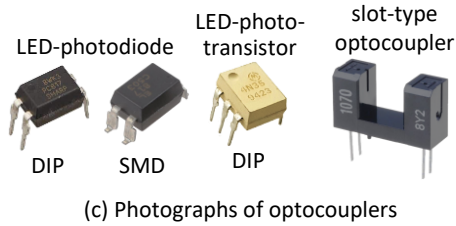


Fig.4-49: Schematic diagrams of optocouplers:
(a) LED-photodiode, (b) LED-phototransistor



Optocouplers are used to isolate sections of a circuit that are incompatible in terms of the voltage levels or currents required. For example, they are used to protect hospital patients from shock when they are connected to monitoring instruments or other devices. They are also used to isolate low-current control or signal circuits from noisy power supply circuits or higher-current motor and machine circuits.

Fig.4-49(c) shows a slot-type optocoupler. The package has a physical gap between the LED and the optodiode or optotransistor. Normally, the LED is turned ON, so the optotransistor always gets light. When an object passes through the gap, it interrupts the light, and the controlling circuit can detect it. Some important applications of this device are- in robotics, speed sensing, printers, and motor drivers, etc.

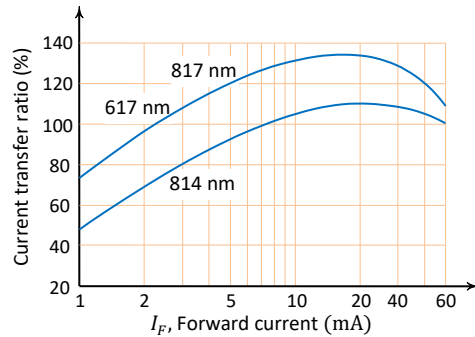


Fig.4-50: Spectral response of optocouplers
(617 nm, 814 nm, and 817 nm wavelength)

Example 4-15

Fig.4-51 shows a switching circuit where the input and output sides are isolated using optocoupler. Calculate the value of R_1 so that the output voltage (V_{CE}) becomes zero / V_{CC} . Assume that the CTR of the optocoupler is 120% (minimum) and the forward voltage drop of the LED is 1.4 V (maximum).

Solution:

As shown in the Fig.4-51, the output voltage will be $V_{CC} = 12\text{ V}$, when $v_{in} = 0$. This state of the output will not depend on the value of R_1 , as $v_{in} = 0$.

On the other hand, output will be zero, when $v_{in} = 5\text{ V}$. For this state, the output current should be,

$$I_{out} = \frac{V_{CC} - V_{CE}}{R_2} = \frac{12\text{ V} - 0\text{ V}}{3.3\text{ k}\Omega} \approx 3.64\text{ mA}$$

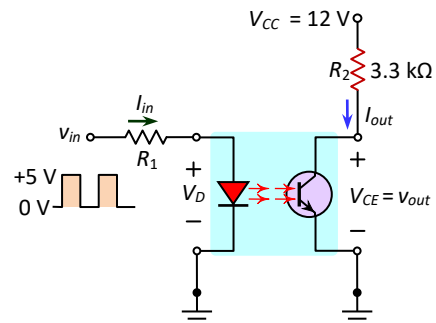


Fig.4-51: Optocoupler circuit
for Example 4-15

Using Equ.(4-37),
$$I_{in} = \frac{I_{out}}{CTR} = \frac{3.64 \text{ mA}}{1.2} \approx 3.03 \text{ mA}$$

From the input circuit we can write, $v_{in} = I_{in}R_1 + V_D$

or,
$$R_1 = \frac{v_{in} - V_D}{I_{in}} = \frac{5 \text{ V} - 1.4 \text{ V}}{3.03 \text{ mA}} \approx 1.19 \text{ k}\Omega \text{ [use standard } 1.2 \text{ k}\Omega\text{] [Ans.]}$$

Comments: Here, the input and output circuit is fully isolated. The transistor inside the optocoupler will work either in cutoff ($V_{CE} \approx 12 \text{ V}$) or in saturation ($V_{CE} \approx 0.2 \text{ V}$), here it is considered as 0 V.

4.19 Sample Questions:

1. Why BJT is named so? Describe the construction of transistors.
2. Describe the transistor action for an NPN transistor or describe the operation of an NPN transistor.
3. What is current gain of a transistor in CBC? Derive its expression.
4. What is I_{CBO} ? Explain with a diagram. Find the expression of I_E in terms of α and I_{CBO} .
5. Explain how a transistor amplifies a DC signal.
6. Describe how a transistor amplifies an AC signal.
7. Describe different configurations of transistors.
8. Describe the input and output characteristics of a transistor in CBC.
9. Describe the input and output characteristics of a transistor in CEC.
10. Describe the transfer characteristics of a transistor in CBC.
11. Describe the transfer characteristics of a transistor in CEC.
12. Give a comparison between the input characteristics of a transistor in CBC and CEC.
13. Give a comparison between the output characteristics of a transistor in CBC and CEC.
14. Why the output characteristics of CEC increases rapidly than that of CBC.
15. What is saturation region of CBC to the left-side of y-axis but for the CEC it is to the right?
16. Why the cutoff region of CEC is higher than that of CBC?
17. Explain different breakdown mechanisms of B-C junction of a transistor.
18. What is β ? Find the relationship between the α and β .
19. Derive the exact expression for the collector current I_C for CEC.
20. What is Early effect and Early voltage? Explain with diagram.
21. Describe the input and output characteristics of a transistor in CCC.
22. Describe the transfer characteristics of a transistor in CC configuration.
23. Explain why the input characteristics in CCC are different for the other two characteristics.
24. Why the leakage current in CCC is exactly same as the leakage current in CEC?
25. What is current gain in CCC configuration? Derive the expression for I_E using current gain and leakage current in CC configuration.
26. Find out the relation between the β and γ .
27. Draw the maximum power dissipation curve of a transistor and explain BV_{CEO} .
28. Explain from data sheet V_{CEO} , I_{CM} , $R_{th j-mb}$, V_{CEsat} , and f_T of a transistor.
29. What TO-number is very common in your electronics lab? Why the collector of some transistors are connected to metal body?
30. What is the pinout of BC547, 2N2222, TIP50, TIP4055 transistors?

31. Explain the on-circuit testing process of a transistor?
32. Explain how the h_{FE} of a transistor can be measured using a multimeter.

4.20 Exercise:

1. An NPN transistor is carrying 1000 electrons per second through the emitter. If 10 electrons recombine with holes in base and other electrons pass to the collector, calculate the value of α of the transistor. [Answer: 0.99]
2. Considering the α of Exercise no. 1, and the AC resistance of the $B-E$ junction is $r_{ac} = 50 \Omega$, calculate the AC output voltage of the circuit of Fig.4-7. Also calculate the voltage gain (v_o/v_i). [Answer: $v_o = 0.495 \text{ V pp} = 99$]

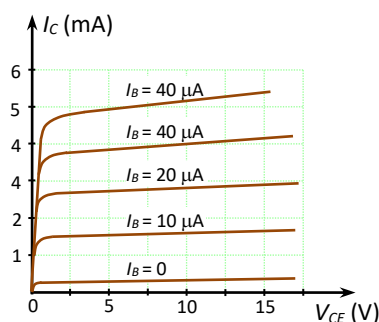


Fig. 4.49: Graph for exercise number 4

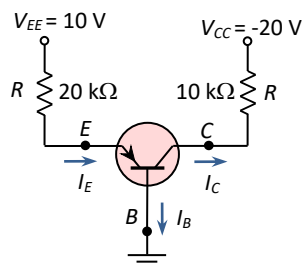


Fig. 4.40 : Circuit for problem no 4

3. From the output characteristic curves given in Fig.4-49 for a CE configuration transistor calculate the value of β_{DC} for $V_{CB} = 5 \text{ V}$ and $V_{CB} = 15 \text{ V}$ and $I_B = 40 \mu\text{A}$. Also calculate their corresponding α_{DC} values. [Answer: $\beta_{dc} \approx 128$; $\beta_{DC} \approx 149$ and $\alpha_{dc} \approx 0.9924$; $\alpha_{dc} \approx 0.9929$.]
4. For the common base circuit shown in Fig. 4.40 calculate: (i) I_B , I_C , and I_E (ii) V_{CE} , V_{CB} , V_E , and V_B . Assume $V_{BE} = 0.7 \text{ V}$, $I_{CBO} = 5 \mu\text{A}$, and $\beta = 75$. [Answer:]

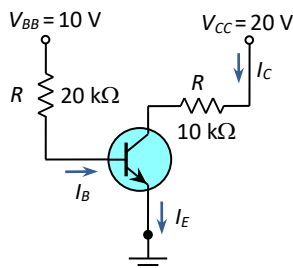


Fig. 4.41 : Circuit for Exercise no. 5

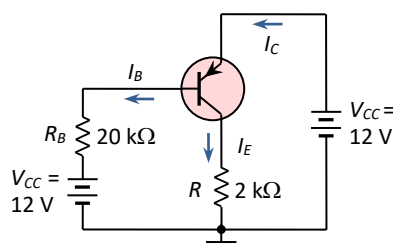


Fig. 4.42 : Circuit for Exercise no. 6

5. Repeat problem number 4 for the common-emitter circuit of Fig. 4.41. [Answer:]

6. Repeat problem number 4 for the common-collector circuit of Fig. 4.42.
7. We have studied that the common base current gain (α) has a minimum value of 0.98 to a maximum value of 0.997. For these values of α , calculate the values of β .
8. We know that the reverse saturation current of a PN junction doubles for every 10 °C rise in temperature. The value of I_{CBO} of a silicon transistor at 25 °C is 1 μ A. If the value of β of the transistor is 150, calculate the value of I_{CBO} and I_{CEO} at 45 °C temperature.
9. From the graphs of Fig. 4.12, determine the value of input DC resistance at $V_{BE} = 0.75$ V of the graph for $V_{CB} = 2$ V.
10. From the graphs of Fig. 4.20., determine the value of input DC resistance at $V_{BE} = 0.75$ V for $V_{CE} = 20$ V.
11. We know that the forward voltage drop of a PN junction doubles for every 10 °C rise in temperature. The value of I_{CBO} of a germanium transistor at 25 °C. If the value of β of the transistor is 150, calculate the values of I_{CBO} and I_{CEO} at 45 °C temperature.
12. A certain transistor has a maximum power rating of ($P_{D(max)} =$) 10 W. The derating factor of this transistor is 12 mW/°C. What will be the derated $P_{D(max)}$ of this transistor at 100°C.

[<https://www.circuitbread.com/study-guides/basic-electronics/bipolar-junction-transistor-bjt-basics>]

13. The common emitter current gain of a transistor is $\beta = 75$. Determine the common base current gain (α).
14. The common base current gain of a transistor is $\alpha = 0.995$. Determine the common emitter current gain (β).
15. The emitter current in a PNP transistor biased in active region is $I_E = 1.50$ mA. The common base current gain of the transistor is $\alpha = 992$. Calculate the value of β , I_B , and I_C .
16. An NPN transistor is biased in the forward-active mode. The base current is $I_B = 20$ μ A and the emitter current is $I_E = 1.8$ mA. Determine the values of α , β , and I_C .

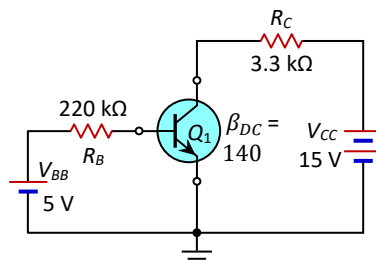


Fig. 4.q : Circuit for Exercise no. 6

17. Considering a Si transistor in the circuit of Fig.4-q, determine the value of I_B , I_C , and V_{CE} .
18. Repeat the above problem considering an ideal transistor ($V_{BE} = 0$).

	$I = \frac{E}{R}$	
or,	$I = \frac{E}{R}$	
	$I = \frac{E}{R}$	(4-38)
or,	$I = \frac{E}{R}$	(4-39)

Example 4-16

Solution:

Comments:

Hj
